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Master Thesis

Gripping Strategy and Parameter Determination Using the PPR-SC Model for Modular Battery Assembly Automation

by

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Abstract

Robotic pick-and-place operations are fundamental to automated manufacturing systems and are widely used in assembly, logistics, and material handling processes. The successful execution of such tasks depends on the accurate determination of gripping parameters, including the grasp location, gripping force, and associated motion configurations. In many industrial applications, these parameters are manually defined by experts or determined through iterative testing in simulation environments. Such approaches are time-consuming and limit the scalability of robotic systems in flexible production scenarios where product variants frequently change. Consequently, automated methods for deriving gripping parameters directly from digital product models have become an important research topic in robotic manipulation.

This thesis proposes a systematic framework for the estimation and validation of gripping parameters based on three-dimensional object models. The methodology integrates geometric data extraction, analytical mechanical modeling, knowledge-based representation, and physics-based simulation. First, relevant geometric and physical properties of an object are extracted from a CAD model using the FreeCAD platform. Based on these properties, analytical formulations derived from classical mechanics are applied to determine candidate gripping points and the minimum gripping forces required to ensure stable manipulation under frictional contact conditions.

To enable structured reasoning and automated configuration of manipulation tasks, the extracted information is organized within a Product–Process–Resource Skill Capability (PPR-SC) framework. The resulting gripping parameters are stored in a JSON-based data structure, which serves as an interface between the planning stage and the simulation environment. The calculated parameters are subsequently validated using a physics-based simulation implemented in MuJoCo, where a robotic manipulator and gripper model interact with the target object under realistic physical constraints.

The results demonstrate that the proposed approach enables the automated generation and verification of gripping parameters using only geometric object information and simplified mechanical models. By combining analytical parameter estimation with ontology-based knowledge representation and simulation-based validation, this work contributes to the development of more autonomous robotic systems capable of configuring pick-and-place operations in flexible manufacturing environments.

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1 Introduction

1.1 Motivation and Industrial Context

The rapid advancement of industrial automation and digital manufacturing has significantly reshaped the role of robotic systems in modern production environments. Contemporary manufacturing is increasingly characterized by high product variability, reduced batch sizes, shorter development cycles, and growing demand for customization. Under these conditions, rigid and preconfigured automation systems are no longer sufficient. Instead, robotic systems must exhibit flexibility, adaptability, and integration within digital engineering workflows. These developments have been widely recognized in recent research on smart industrial robot control and flexible manufacturing systems [1].

Pick-and-place operations constitute one of the most fundamental tasks in industrial robotics. They form the basis of assembly operations, packaging systems, material handling processes, and logistics automation. Although such tasks are often considered elementary, their reliable execution depends on precise configuration of gripping and motion parameters. Inadequate parameterization may lead to object slippage, excessive deformation, misalignment, or even mechanical damage to the product or the robot. The importance of accurate grasp configuration and force estimation in robotic manipulation has been extensively studied in the literature on robotic grasping and contact mechanics [2], [3].

Traditionally, gripping parameters are determined manually by experienced engineers. This process involves selecting a gripping location based on geometric intuition, assigning gripping force based on empirical safety margins, and tuning approach and retreat trajectories through trial and error. While this approach is effective in stable and repetitive production systems, it presents significant limitations in highly dynamic manufacturing environments.

First, manual configuration lacks scalability. Every new product geometry requires renewed analysis and parameter tuning. Second, empirical approaches depend heavily on expert knowledge, which introduces subjectivity and variability. Third, iterative simulation-based adjustment, although more systematic, demands computational effort and detailed modeling expertise. In many industrial cases, the configuration effort exceeds the time required for actual task execution.

At the same time, the increasing adoption of digital product models offers new possibilities for automation. Modern **CAD (Computer Aided Design)** systems provide accurate geometric representations of objects, including dimensional properties, mass distributions, and structural information. Such digital representations contain rich data that can potentially be exploited for automated parameter derivation. CAD-based manipulation planning approaches have demonstrated that object geometry extracted from digital models can be used to determine feasible grasp locations and motion strategies for robotic manipulation tasks [4]. If gripping parameters can be analytically calculated directly from CAD-based geometry, the dependency on manual tuning could be significantly reduced. Analytical models derived from mechanical principles allow the estimation of grasp stability and required gripping forces without relying on purely empirical methods. Such analytical approaches have been explored in grasp quality analysis and robotic manipulation research [5].

Furthermore, the emergence of digital twins and structured data integration in manufacturing systems emphasizes the importance of linking product data, process requirements, and resource capabilities in a unified framework. In this context, ontology-based knowledge representation enables formal modeling of relationships between objects, tasks, and resources. Such structured representation allows automated reasoning and supports intelligent configurations of robotic systems in flexible production environments [1].

The motivation of this thesis therefore arises from the convergence of three developments:

- The need for scalable and systematic gripping parameter estimation.
- The availability of detailed digital product models.
- The potential of structured knowledge representation and analytical modeling.

By integrating these elements, robotic manipulation can move from empirically tuned configurations toward analytically derived and digitally validated parameterization workflows.

1.2 Problem Definition and Objectives

In robotic manipulation tasks, gripping parameters define the interface between the robot and the manipulated object. These parameters determine whether the grasp is stable, whether the object can be transported without slipping, and whether placement occurs within acceptable tolerances. The importance of accurate grasp configuration and contact stability has been widely discussed in the literature on robotic grasping and manipulation mechanics [2], [5].

The essential parameters in a pick-and-place task include:

- The spatial location of the gripping point relative to the object's geometry.
- The orientation of the gripper with respect to the object.
- The magnitude of the gripping force required to prevent translational and rotational slip.
- The pre-pick and post-pick approach positions.
- The release height and final placement configuration.

The determination of these parameters involves physical considerations such as static equilibrium, friction conditions, torque balance, and geometric accessibility. These principles form the foundation of classical robotic grasp analysis and manipulation mechanics [3], [5]. However, existing approaches often treat these considerations implicitly rather than explicitly.

High-fidelity simulation environments provide accurate modeling of contact forces, friction behaviour, and dynamic motion. Such simulation tools allow detailed evaluation of robotic manipulation tasks and have become increasingly important for validating grasp strategies and motion planning algorithms [6], [7]. While powerful, such tools typically serve as validation mechanisms rather than parameter derivation methodologies. Moreover, they require detailed configuration and do not inherently explain the physical reasoning behind selected parameters.

On the other hand, purely analytical approaches rely on mechanical equations describing force equilibrium and friction constraints. For instance, the prevention of slipping requires that the tangential forces remain below the friction limit, and rotational stability requires torque balance around the centre of mass. Analytical methods offer transparency and computational efficiency but may neglect practical influences such as geometric tolerances, manufacturing deviations, and uncertainties in material properties. The relationship between grasp stability and friction constraints has been extensively analyzed in the context of grasp quality metrics and contact mechanics [5], [8].

In addition to mechanical considerations, gripping parameter estimation requires integration of contextual information. The gripping force must not exceed the mechanical limits of the gripper. The pre-pick position must lie within the robot's reachable workspace. The release height must account for process constraints. Therefore, gripping parameter estimation is not solely a geometric or mechanical problem, but rather a multi-domain integration problem involving product, process, and resource data. Structured modeling approaches have been proposed to support such integration in advanced manufacturing systems and intelligent robotic automation frameworks [1].

The central challenge addressed in this thesis is thus not merely the calculation of a gripping force or position, but the development of a structured methodology that integrates:

- Geometric information from CAD models,
- Mechanical principles for stability analysis,
- Structured knowledge representation for resource configuration,
- Physics-based simulation for validation.

The overarching research problem can be expressed as *"How can gripping parameters for robotic pick-and-place operations be systematically derived from digital product models using analytical methods and structured knowledge representation, and how reliable are these parameters when validated through physics-based simulation?"*

To address this problem, the following objectives are defined:

1. Develop a coherent framework for automated gripping parameter estimation.
2. Identify and formalize the necessary data categories required for parameter derivation.
3. Establish analytical models for calculating gripping forces and stability conditions.
4. Integrate ontology-based knowledge modeling to support automated reasoning and resource configuration.
5. Validate the analytically derived parameters in a MuJoCo-based physics simulation environment.

By pursuing these objectives, this thesis aims to demonstrate that gripping parameter estimation can transition from empirical configuration toward analytically grounded and digitally integrated workflows.

1.3 Scope and Structure of the Thesis

This thesis focuses on rigid-body pick-and-place tasks executed with a parallel-jaw gripper under quasi-static conditions. Parallel-jaw grippers represent one of the most widely used end-effector configurations in industrial robotic manipulation due to their mechanical simplicity and robustness in structured manufacturing environments [9], [10]. The object is assumed to be represented by a detailed 3D CAD model from which geometric properties and mass characteristics can be extracted. CAD-based manipulation planning approaches have demonstrated that geometric information obtained from digital object models can be effectively used for determining feasible grasp locations and manipulation strategies [4]. Friction coefficients and material parameters are assumed to be known or estimated within reasonable bounds.

The analysis concentrates on stable two-point gripping scenarios and does not extend to complex multi-finger dexterous manipulation. Deformable object handling, soft robotic grippers, and real-time sensor-based adaptive grasping are beyond the scope of this study. These topics represent active research areas within robotic manipulation but introduce additional complexities related to compliance modelling, tactile sensing, and advanced control strategies [11], [12].

Additionally, real-time computer vision-based grasp detection is not considered; instead, the emphasis lies on geometry-driven parameter estimation from digital models. These topics represent active research areas within robotic manipulation but introduce additional complexities related to compliance modelling, tactile sensing, and advanced control strategies [11], [12].

The methodological scope includes:

- Extraction of geometric and physical properties from CAD models using FreeCAD.
- Analytical derivation of gripping force and stability conditions.
- Definition of a JSON-based interface for structured data exchange.
- Physics-based validation using MuJoCo simulation.

2 Theoretical Foundations and State of the Art

2.1 Principles of Robotic Grasping and Manipulation

Grasping involves establishing controlled contact forces between a robotic end-effector and an object such that the object can be securely held and transported under external disturbances. As emphasized in foundational work on robotic grasping [2], the mechanical interaction between the gripper and the object forms the basis of all higher-level manipulation tasks.

From a mechanical perspective, grasping can be divided into two fundamental functions: restraining (or fixturing) and skillful manipulation [2]. In industrial pick-and-place applications, the primary objective is restraining, ensuring that the object remains stable during motion of the robot arm. Parallel-jaw grippers are widely used in such applications due to their simplicity, repeatability, and robustness [9]. Despite their mechanical simplicity, the physical phenomena governing grasp stability are complex and involve contact mechanics, frictional interactions, and force distribution.

A grasp is commonly modelled as a set of contact points between the gripper and the object. Each contact generates a force vector that can be decomposed into a normal component and a tangential component. Under Coulomb friction assumptions, the tangential force is bounded by the normal force scaled by the friction coefficient. The admissible set of contact forces is geometrically represented by the friction cone, a concept extensively discussed in grasp mechanics literature [2], [5]. If the resultant contact force lies within the friction cone, slipping is prevented; otherwise, the object will slide relative to the contact surface.

One of the most important theoretical concepts in grasp analysis is force closure. A grasp achieves force closure if the set of contact forces can counteract any external wrench (combined force and torque) applied to the object [2], [5]. The mathematical framework for analysing force closure often relies on the grasp wrench space, which represents the set of wrenches that can be generated by the contact forces. Ferrari and Canny introduced quantitative grasp quality measures based on the distance to the boundary of the wrench space, enabling evaluation of robustness against disturbances [8]. Later reviews, such as that by **Roa and Suárez**, systematically categorized grasp quality metrics and evaluated their computational properties and industrial applicability [5].

Although these formulations are mathematically rigorous, industrial pick-and-place tasks typically operate under structured conditions where gravitational forces dominate external disturbances. In such scenarios, grasp stability primarily depends on sufficient frictional force to balance the object's weight and sufficient moment resistance to prevent rotational instability. As discussed in the Springer Handbook of Robotics [3], classical grasp analysis provides the theoretical foundation for these considerations.

The development of grasp planning methods can broadly be divided into analytical, model-based approaches and data-driven approaches. Analytical approaches use geometric and mechanical models of the object to derive grasp configurations. Early work focused on multi-fingered hands and skillful manipulation [2], whereas later research addressed practical industrial grippers and structured environments. For example, Wan et al. proposed model-based grasp planning algorithms for assembly tasks using CAD models to synthesize stable grasp configurations for industrial end-effectors [13]. Similarly, **Kleeberger et al.** presented an automatic offline grasp pose generation method for parallel-jaw grippers, incorporating geometric constraints, friction considerations, and collision checking [10]. These approaches demonstrate that when accurate CAD models are available, grasp poses can be systematically derived without relying on empirical trial-and-error methods.

In industrial contexts, gripper dimensioning and configuration are often determined based on object properties such as mass, geometry, and surface characteristics. **Schmalz et al.** presented an automated method for dimensioning gripper systems based on object parameters and process requirements [14], highlighting the importance of integrating product data into gripping system design. Surveys of industrial grippers further emphasize that gripping performance depends not only on geometry but also on material properties, surface roughness, and compliance characteristics [9].

In parallel with analytical developments, data-driven and learning-based grasp planning has gained significant attention. Dex-Net introduced a cloud-based framework for robust grasp planning using large datasets of 3D object models and probabilistic force-closure estimation under uncertainty [15]. Vision-based grasping surveys demonstrate how deep learning methods can estimate grasp poses directly from RGB-D data without explicit geometric modeling [16]. More recent work explores physics-based self-supervised learning approaches that combine simulation and real-world data to improve grasp robustness [17].

Affordance based methods further attempt to generalize grasp detection across object categories without relying on object identity [18].

While learning-based approaches have shown impressive generalization capabilities in unstructured environments, they typically require extensive training data and computational resources. Furthermore, their decision-making processes are often less interpretable than analytical methods. In highly structured industrial environments, where object geometry is known and CAD models are available, analytical and model-based approaches remain highly relevant [10], [13]. In such settings, deterministic parameter derivation based on mechanical principles can provide transparency, reproducibility, and computational efficiency.

Recent surveys on high-precision robotic grasping and assembly tasks emphasize the growing need for integration between perception, mechanics, and control [11]. Smart industrial robot control strategies increasingly combine model-based reasoning with learning-based components to achieve flexibility while maintaining reliability [1]. Nevertheless, for structured pick-and-place tasks involving rigid objects and parallel-jaw grippers, the dominant stability constraints remain fundamentally mechanical.

Despite the maturity of grasp theory, several challenges remain when transferring theoretical formulations to industrial practice. Classical models assume rigid bodies, precise contact locations, and uniform friction distribution [2], [5]. In real manufacturing environments, geometric tolerances, surface variations, and material compliance may alter contact conditions. As a result, practical implementations often incorporate safety factors or simulation-based validation to ensure robustness.

The literature thus reveals a spectrum of approaches ranging from purely analytical grasp synthesis to fully data-driven methods. In structured industrial scenarios with available CAD data, model-based analytical methods provide a transparent and computationally efficient foundation for parameter estimation. At the same time, insights from simulation-based and learning-based research highlight the importance of accounting for uncertainty and variability.

For the purposes of this thesis, grasping is treated primarily as a mechanically constrained restraining problem under quasi-static conditions. The focus lies on deriving gripping parameters, such as gripping force and gripping location, from geometric and physical object properties. The theoretical principles summarized in this section establish the foundation for the analytical parameter estimation framework developed in subsequent chapters.

2.2 Analytical Mechanics of Gripping and Stability

The stability of a robotic grasp is fundamentally governed by rigid body mechanics and contact force interactions. While Section 2.1 established the conceptual framework of grasping and force closure, the present section develops the analytical foundations required for systematic gripping parameter estimation in structured pick-and-place tasks. In industrial applications involving parallel-jaw grippers, grasp stability is typically evaluated under quasi-static conditions where gravitational loading dominates external disturbances [3].

For a rigid body in static equilibrium, Newton–Euler conditions require that the sum of forces (F) and the sum of moments (M) acting on the body neutralize:

$$\sum F = 0 \tag{2.1}$$

$$\sum M = 0 \tag{2.2}$$

where forces and moments are evaluated with respect to a reference point, typically the center of mass. These equilibrium conditions form the mechanical basis for evaluating whether a given gripping configuration can restrain an object during lifting and transport.

In a vertical pick-and-place operation, the dominant external force is gravitational force:

$$F_g = mg \tag{2.3}$$

where **m** denotes mass of the object and **g** denotes the gravitational acceleration.

In a symmetric two-finger parallel-jaw grasp, each finger applies a normal force $\boldsymbol{F_n}$ perpendicular to the contact surface. The maximum available tangential (frictional) force $\boldsymbol{F_t}$ at each contact is limited by Coulomb friction:

$$|F_t| \leq \mu F_n \tag{2.4}$$

where $\boldsymbol{\mu}$ is the coefficient of friction between gripper and object and this value depends on the materials of the object and the gripper. The friction cone model, widely discussed in classical grasp theory [2], [5], describes the admissible force region preventing slip.

To prevent translational slip during lifting, the sum of frictional forces from both the fingers must exceed gravitational force:

$$2\mu F_n \geq mg \tag{2.5}$$

This condition yields the minimum required normal force per finger:

$$F_n \geq \frac{mg}{2\mu} \tag{2.6}$$

This expression is frequently used in industrial gripper dimensioning and force selection procedures [14]. However, translational equilibrium alone does not ensure grasp stability. If the line of action of the gripping forces does not pass through the center of mass, gravitational loading generates a torque:

$$M_g = mgd \tag{2.7}$$

where **d** is the perpendicular distance between the gripping line and the center of mass. This torque must be counteracted by frictional moments generated at the contact interfaces.

Assuming symmetric contacts separated by an effective lever arm **2r**, the maximum resisting frictional moment is:

$$M_f = 2\mu F_n r \tag{2.8}$$

Rotational equilibrium therefore requires:

$$2\mu F_n r \geq mgd \tag{2.9}$$

Solving for the normal force yields:

$$F_n \geq \frac{mgd}{2\mu r} \tag{2.10}$$

This relationship highlights several important dependencies. First, the required gripping force increases proportionally with object mass and center-of-mass offset. Second, higher friction coefficients reduce required force. Third, increasing contact spacing improves rotational stability. These dependencies align with industrial observations that gripping near the center of mass reduces required gripping force and increases robustness [10], [13].

Combining translational and rotational constraints results in a unified stability condition:

$$F_{required} \geq \frac{mg}{2\mu} \max\left(1, \frac{d}{r}\right) \tag{2.11}$$

This expression provides a deterministic analytical mapping from object properties like mass, center-of-mass location, and friction coefficient, to required gripping force. Such explicit formulations are advantageous in structured environments where CAD models allow accurate extraction of geometric parameters [13].

However, real industrial systems rarely satisfy idealized modeling assumptions. Classical analytical formulations assume rigid bodies, exact contact locations, and constant friction coefficients [2], [5]. In practice, geometric tolerances and surface variations alter contact conditions. Manufacturing tolerances may shift the center of mass by a deviation Δ d, modifying the gravitational torque:

$$M_g = mg(d + \Delta d) \tag{2.12}$$

where **d** represents the nominal distance between the gripping point and the center of mass, and Δd accounts for possible positional deviations due to manufacturing tolerances.

Similarly, small misalignments in gripper positioning may reduce effective lever arm r, increasing required gripping force. As discussed in high-precision assembly literature [11], such uncertainties can significantly influence stability margins.

Material compliance further complicates contact modeling. Although rigid body assumptions simplify analysis, real contact interfaces exhibit elastic deformation. Elastic deformation alters pressure distribution and effective contact area, potentially increasing or decreasing frictional capacity depending on surface characteristics [9]. Advanced modeling of such effects typically requires finite element analysis or contact mechanics simulations, which exceed the scope of purely analytical derivation

Dynamic effects must also be considered. Although many pick-and-place operations are designed to operate under quasi-static conditions, acceleration during robot motion introduces inertial forces:

$$F_{inertial} = ma \tag{2.13}$$

Under vertical acceleration, the effective load becomes:

$$F_{effective} = m(g + a) \tag{2.14}$$

Consequently, required gripping force increases with acceleration magnitude. Intelligent gripper control strategies aim to adapt gripping force dynamically to prevent slip under varying loads [19]. However, in many industrial systems, gripping force is selected conservatively using safety factors.

To account for uncertainties in friction coefficients, geometric tolerances, and dynamic loading, safety factors S_f are introduced:

$$F_{design} = S_f F_{required} \tag{2.15}$$

where $S_f > 1$ compensates for modeling inaccuracies. The selection of appropriate safety factors remains largely empirical in industry [14], reinforcing the need for structured analytical reasoning combined with validation.

While learning-based grasp planning methods, such as Dex-Net, incorporate probabilistic force closure estimation under uncertainty [15], their underlying mechanical stability criteria remain rooted in classical friction and equilibrium analysis. Even data-driven approaches ultimately evaluate grasp stability using mechanical metrics derived from contact mechanics [5], [8].

The analytical relationships developed in this section form the mechanical core of the gripping parameter estimation framework proposed in this thesis. By extracting mass properties and geometric information from CAD models, it becomes possible to compute required gripping force and evaluate stability conditions deterministically. However, given the simplifying assumptions inherent in analytical modeling, simulation-based validation is introduced later to verify stability under more realistic physical conditions.

In summary, analytical mechanics provides a transparent and computationally efficient foundation for gripping parameter estimation. The derived equations establish explicit relationships between object geometry, mass distribution, friction properties, and required gripping force. These relationships enable systematic parameter generation directly from digital product models, forming a key pillar of the overall methodology.

2.3 Ontology-Based Knowledge Representation in Robotic Systems

While analytical mechanics provides the physical foundation for grasp stability, gripping parameter estimation in industrial environments is not solely a mechanical problem. It requires the integration of heterogeneous information sources, including product properties, process constraints, and resource capabilities. In modern manufacturing systems, such integration is increasingly addressed through structured knowledge representation and semantic modeling techniques.

Ontology-based knowledge representation provides a formal framework for modeling entities, relationships, and constraints within a domain. An ontology defines classes, properties, and logical relations in a machine-interpretable format, enabling automated reasoning and consistency checking. In robotics and manufacturing contexts, ontologies are frequently employed to formalize the relationships between products, processes, and resources, facilitating interoperability and intelligent system configuration [1].

The Product–Process–Resource (PPR) paradigm is widely adopted in digital manufacturing environments. Within this paradigm, the product describes geometric and physical properties of the workpiece, the process defines task-related constraints and goals, and the resource represents the capabilities and limitations of the robotic system and gripper. As industrial automation systems become increasingly flexible and digitally integrated, structured modeling of PPR relationships becomes essential for autonomous configuration and planning [1].

In the context of robotic grasping, product information includes mass, center-of-mass location, geometry, surface characteristics, and material properties. Process information includes required placement accuracy, release height, allowable orientation deviations, and cycle time constraints. Resource information encompasses gripper stroke limits, maximum gripping force, reachable workspace, and sensor capabilities. These heterogeneous data categories must be coherently combined to derive valid gripping parameters.

Traditional robotic control architectures often embed such relationships implicitly within software logic. However, implicit encoding limits transparency and reusability. Ontology-based modeling, in contrast, externalizes domain knowledge and enables formal reasoning over constraints. For example, if the analytically computed gripping force exceeds the maximum allowable force of the gripper, ontology-based reasoning can automatically identify infeasibility and trigger alternative configuration strategies. Similarly, if the required gripper opening exceeds the stroke limit, the system can detect constraint violations before execution.

The increasing complexity of robotic manipulation systems has motivated research into intelligent and cognitive robot control architectures [1]. Smart industrial robot control strategies emphasize integration of perception, reasoning, and actuation within unified digital frameworks. Ontologies play a crucial role in such architectures by providing semantic interoperability between modules and enabling reasoning across different abstraction levels.

In grasp planning literature, knowledge representation has often been implicit. For instance, CAD-based grasp planning methods utilize geometric segmentation and surface classification to generate grasp candidates [10], [13]. Although these approaches are model-based, the decision logic linking geometric properties to grasp feasibility is typically embedded algorithmically rather than represented semantically. By contrast, ontology-based approaches allow explicit modeling of relationships such as:

- Object mass influences required gripping force.
- Required gripping force must not exceed gripper capacity.
- Gripper type determines allowable contact configurations.

Such formalization enhances modularity and traceability of parameter estimation.

Data-driven grasp planning methods, including Dex-Net [15] and vision-based grasp estimation techniques [16], often rely on large datasets and probabilistic evaluation of grasp success. While powerful, these approaches do not explicitly represent process or resource constraints beyond learned statistical patterns. In industrial contexts, however, compliance with hard constraints—such as maximum force limits or workspace boundaries—is mandatory. Ontology-based modeling complements analytical and learning-based approaches by enforcing structured constraint handling.

In assembly and regrasp planning research, integration of task-level reasoning and motion planning has been addressed through graph-based representations [6], [20]. These methods illustrate the importance of structured representation for managing complex manipulation sequences. Similarly, high-precision assembly surveys emphasize the need to integrate sensing, control, and mechanical constraints within coherent frameworks [11].Ontology-based representation offers a formal mechanism for achieving such integration.

From a theoretical perspective, ontologies are commonly implemented using description logics and standards such as the Web Ontology Language (OWL). These formalisms support class hierarchies, property restrictions, and logical inference. While the present thesis does not aim to develop a complete semantic web framework, the conceptual adoption of ontology principles supports structured representation of gripping knowledge. By modeling PPR entities explicitly, it becomes possible to:

- Automatically verify feasibility of analytically derived parameters.
- Ensure consistency between product properties and resource limits.
- Support extensibility for additional object types or gripper configurations.

The integration of ontology with analytical mechanics yields several advantages. Analytical equations provide quantitative parameter estimation, while ontology-based reasoning ensures qualitative constraint satisfaction. This combination bridges low-level mechanical modeling and high-level system configuration.

Furthermore, in digitally integrated production environments aligned with Industry 4.0 principles, semantic interoperability between systems is increasingly important [1]. Structured knowledge representation facilitates communication between CAD systems, simulation environments, and robotic controllers. In the workflow proposed in this thesis, gripping parameters derived from CAD models are exported in structured formats (e.g., JSON (JavaScript Object Notation)) and validated in simulation. Ontology-based modeling supports the coherent organization of the underlying knowledge required for such data exchange.

Despite these advantages, ontology-based approaches also introduce challenges. The development of comprehensive ontologies requires careful domain modeling and maintenance. Overly complex ontological structures may increase system overhead without proportional benefit. Therefore, the ontology framework in this thesis is designed to be minimal yet sufficient for representing essential product, process, and resource relationships relevant to gripping parameter estimation.

In summary, ontology-based knowledge representation provides a structured mechanism for integrating heterogeneous data required for gripping parameter estimation. While analytical mechanics determines the quantitative requirements for stable grasping, ontology modeling ensures that these requirements are consistent with process constraints and resource capabilities. The combination of mechanical modeling and structured knowledge representation forms the conceptual backbone of the framework developed in subsequent chapters.

2.4 Simulation-Driven Validation of Manipulation Tasks

While analytical mechanics provides deterministic relationships for gripping force and stability estimation, real-world manipulation involves complex physical interactions that exceed the assumptions of ideal rigid-body models. Contact dynamics, friction variability, geometric tolerances, and dynamic motion introduce uncertainties that may not be fully captured by closed-form analytical equations. For this reason, physics-based simulation plays a critical role in validating analytically derived manipulation parameters before deployment in physical systems.

Physics simulation engines model rigid body dynamics, joint kinematics, collision detection, and frictional contact interactions in a unified computational framework. These environments numerically integrate Newton–Euler equations of motion while resolving contact constraints. In grasping applications, simulation enables evaluation of slip behavior, contact stability, collision avoidance, and motion feasibility under controlled virtual conditions.

The mathematical basis of rigid body simulation is rooted in classical mechanics. The motion of a rigid body is governed by:

$$m\ddot{x} = \sum F \tag{2.16}$$

$$I\dot{\omega} = \sum M \tag{2.17}$$

where $\mathbf{m}$ is mass, $\mathbf{I}$ is the inertia tensor, $\mathbf{x}$ is the translational acceleration and $\boldsymbol{\omega}$ is the angular acceleration. Contact forces are computed through constraint-based or penalty-based formulations. In Coulomb friction models, tangential forces are limited by normal forces scaled by friction coefficients, consistent with the friction cone concepts introduced in Section 2.1.

In the context of robotic grasping, simulation is frequently used to evaluate grasp stability under uncertainty. Dex-Net integrates probabilistic force-closure estimation with physics-based evaluation to quantify grasp robustness under pose and friction variability [15]. More recent work incorporates physics-based self-supervised learning to refine grasp pose detection by simulating contact outcomes [17]. These approaches demonstrate that even learning-based grasp planning relies fundamentally on accurate physical simulation.

Industrial applications of simulation extend beyond grasp synthesis. Simulation models enable verification of pick-and-place operations, motion planning feasibility, and collision avoidance. **Calvo et al.** developed a simulation model for robotic pick-and-place systems to evaluate performance and optimize system configuration [21]. Such work highlights the role of simulation in reducing commissioning time and increasing reliability in industrial automation.

Compared to purely analytical approaches, simulation offers several advantages:

1. **Inclusion of complex contact interactions.**

   Contact geometry may involve multiple points, non-planar surfaces, and varying friction conditions that are difficult to capture analytically.

2. **Modeling of dynamic motion.**

   Acceleration-induced inertial forces can be evaluated directly during simulated trajectories.

3. **Evaluation under uncertainty.**

   Variations in friction coefficients, mass distribution, or geometric tolerances can be incorporated into scenario testing.

4. **Collision detection and reachability analysis.**

   Motion feasibility can be verified within realistic workspace constraints.

However, simulation also has limitations. The accuracy of results depends on correct parameterization of physical properties such as friction coefficients and contact stiffness. Numerical integration methods introduce discretization errors, and computational complexity increases with system size. Furthermore, simulation does not inherently explain why a given parameter configuration succeeds or fails; it serves as a validation tool rather than a derivation mechanism.

In industrial robotics research, simulation environments such as Gazebo, Bullet, and MuJoCo are commonly used for contact-rich manipulation tasks. **MuJoCo** (Multi-Joint dynamics with Contact) has gained prominence due to its efficient handling of articulated systems and stable contact resolution algorithms. It employs constraint-based formulations for contact modeling and supports configurable friction and compliance parameters, making it suitable for grasp stability analysis.

Recent literature demonstrates the integration of simulation into grasp planning workflows. **Ruiz et al**. proposed a physics-based self-supervised grasp pose detection framework, where simulation generates labeled grasp outcomes for learning algorithms [17]. Such approaches emphasize that accurate physical modeling enhances reliability of grasp estimation.

In addition to learning-based validation, simulation supports model-based approaches. CAD-based grasp planning systems, such as those presented by **Wan et al.** [13] and **Kleeberger et al.** [10], typically incorporate collision checking and stability testing within virtual environments before deployment. This layered validation strategy reflects best practices in industrial automation.

From a methodological perspective, simulation complements analytical modeling by relaxing idealized assumptions. For example, while analytical derivations assume rigid bodies and uniform friction, simulation can incorporate:

- Variable friction coefficients across surfaces,
- Slight misalignments in gripper positioning,
- Non-ideal contact geometry,
- Dynamic acceleration profiles,
- Elastic contact approximations.

Simulation thus provides a mechanism for stress-testing analytically derived gripping parameters under more realistic conditions.

In the framework proposed in this thesis, simulation is explicitly positioned as a validation stage, not as the primary parameter generator. Gripping force, gripping location, pre-pick and post-pick poses are first derived analytically using CAD-extracted geometric and physical properties. These parameters are then transferred into a simulation environment to evaluate:

- Translational and rotational stability,
- Slip occurrence,
- Motion feasibility,
- Sensitivity to tolerance variations.,

This separation between derivation and validation enhances methodological clarity. Analytical equations provide transparency and computational efficiency, while simulation verifies robustness against modeling simplifications.

The integration of simulation within digital engineering workflows aligns with Industry 4.0 principles emphasizing digital twins and virtual commissioning [1]. By validating manipulation tasks virtually before physical execution, commissioning time can be reduced and failure risks minimized.

In summary, physics-based simulation constitutes a critical component of modern robotic manipulation systems. While analytical mechanics establishes fundamental stability conditions, simulation enables evaluation under realistic contact and motion scenarios. The combination of analytical derivation and simulation validation provides a balanced methodology that leverages the strengths of both deterministic modeling and numerical evaluation. This integrated perspective forms the basis for the experimental validation strategy developed in later chapters.

2.5 Product-Process-Resource-Skill-Capability Modelling Approach

Modern manufacturing systems increasingly demand flexible and adaptive automation architectures capable of handling frequent changes in product design, production volumes, and assembly processes. Traditional automation systems are typically developed using rigid programming structures where product-specific parameters are embedded directly into robot programs or control logic. While such approaches may function efficiently in fixed production environments, they often lack the flexibility required in modern manufacturing contexts where new product variants and design modifications are introduced regularly. As a result, reconfiguring automation systems can require significant engineering effort and system downtime. To address these challenges, contemporary research in industrial automation has focused on the development of knowledge-based modelling approaches that separate product information from process descriptions and machine capabilities.

One of the most widely used modelling paradigms in manufacturing systems engineering is the **Product-Process-Resource (PPR)** model, which provides a structured framework for representing manufacturing activities by separating production information into three interconnected domains. The product represents the physical item being manufactured or manipulated, the process describes the sequence of operations required to transform or manipulate the product, and the resource refers to the machines, tools, and software systems that execute these processes. By separating these three elements, the PPR paradigm provides a clear and modular representation of manufacturing systems, enabling better reuse of engineering knowledge and improved system configurability. Such modelling approaches have been widely discussed in production systems engineering and digital manufacturing research [22].

The PPR modelling approach is widely used in digital manufacturing environments because it supports the integration of engineering data across different stages of the product lifecycle. In modern production systems, digital product models generated through computer-aided design (CAD) tools serve as the primary source of geometric and physical information about manufactured components. These models can be directly linked to manufacturing processes and production resources through structured modelling frameworks such as PPR. By maintaining this separation between product data, process logic, and resource capabilities, the PPR model facilitates the development of flexible automation systems that can adapt to changes in product design without requiring extensive modifications to the underlying control architecture [4], [22].

Figure 2.1 illustrates the conceptual structure of the PPR modelling paradigm used in manufacturing system design. In this framework, the product entity defines the physical characteristics of the object being manipulated, including its geometric dimensions, mass properties, and material attributes. The process entity defines the sequence of operations required to perform a manufacturing or manipulation task, while the resource entity represents the physical equipment used to execute these operations. By clearly separating these three components, the PPR model enables engineers to describe manufacturing systems in a modular and scalable manner.

Within robotic manipulation systems, the PPR framework provides a useful abstraction for modelling pick-and-place operations. In this context, the manipulated object corresponds to the product entity, the manipulation sequence corresponds to the process entity, and the robotic manipulator together with its gripper represents the resource performing

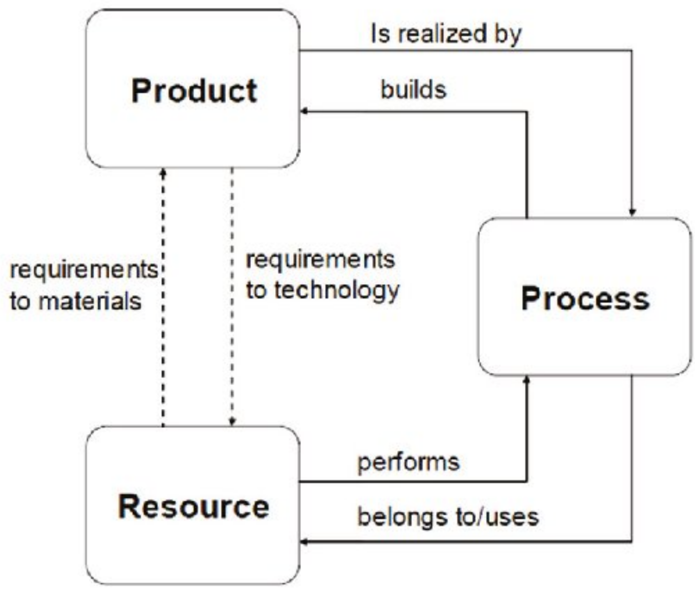

Fig. 2.1: Product-Process-Resource (PPR) workflow.

the operation. For example, when a robotic system is tasked with transferring a battery module from a storage location to an assembly fixture, the battery module itself is represented as the product, the sequence of actions such as approach, grasping, lifting, and placement constitute the process, and the robotic manipulator with its gripper mechanism serves as the resource executing the task.

The PPR modelling paradigm also supports the integration of digital engineering tools within manufacturing workflows. Since product data are typically stored in CAD models, these models can serve as the starting point for automated process planning and parameter estimation. CAD models provide detailed geometric descriptions of objects, including dimensions, surface geometry, and mass distribution, which are essential for determining feasible grasp locations and manipulation strategies. By linking CAD-based product information with process definitions and resource capabilities, the PPR model enables automated reasoning about manufacturing operations and facilitates the development of intelligent automation systems [4].

Despite its advantages, the classical PPR model has certain limitations when applied to modern cyber-physical production systems. In particular, the traditional PPR framework does not explicitly describe the functional abilities of production resources. While the resource layer identifies the machines involved in a manufacturing process, it does not provide a detailed representation of what these machines are capable of performing. This limitation becomes particularly significant in flexible manufacturing environments where multiple machines may be capable of performing similar tasks, and where automated task allocation requires a formal description of machine capabilities.

To address this limitation, recent research in industrial automation has proposed extending the classical PPR framework by introducing additional modelling layers describing skills and capabilities of production resources. In this extended modelling approach, capabilities represent the abstract functional abilities of a resource, while skills correspond to executable operations that implement these capabilities within the control system. For example, a robotic gripper may possess the capability to apply a certain gripping force range, while the associated skill may represent the execution of a grasping operation within a robotic control program.

The integration of these additional layers leads to **Product–Process–Resource -Skill-Capability** modelling framework, which provides a more comprehensive representation of manufacturing systems. In this framework, processes are described in terms of the capabilities required to execute them, while resources are described in terms of the capabilities they possess. Skills act as the executable interface between process requirements and resource capabilities. This modelling approach allows production tasks to be dynamically assigned to resources that possess the necessary capabilities, thereby enabling greater flexibility and adaptability in manufacturing systems [23].

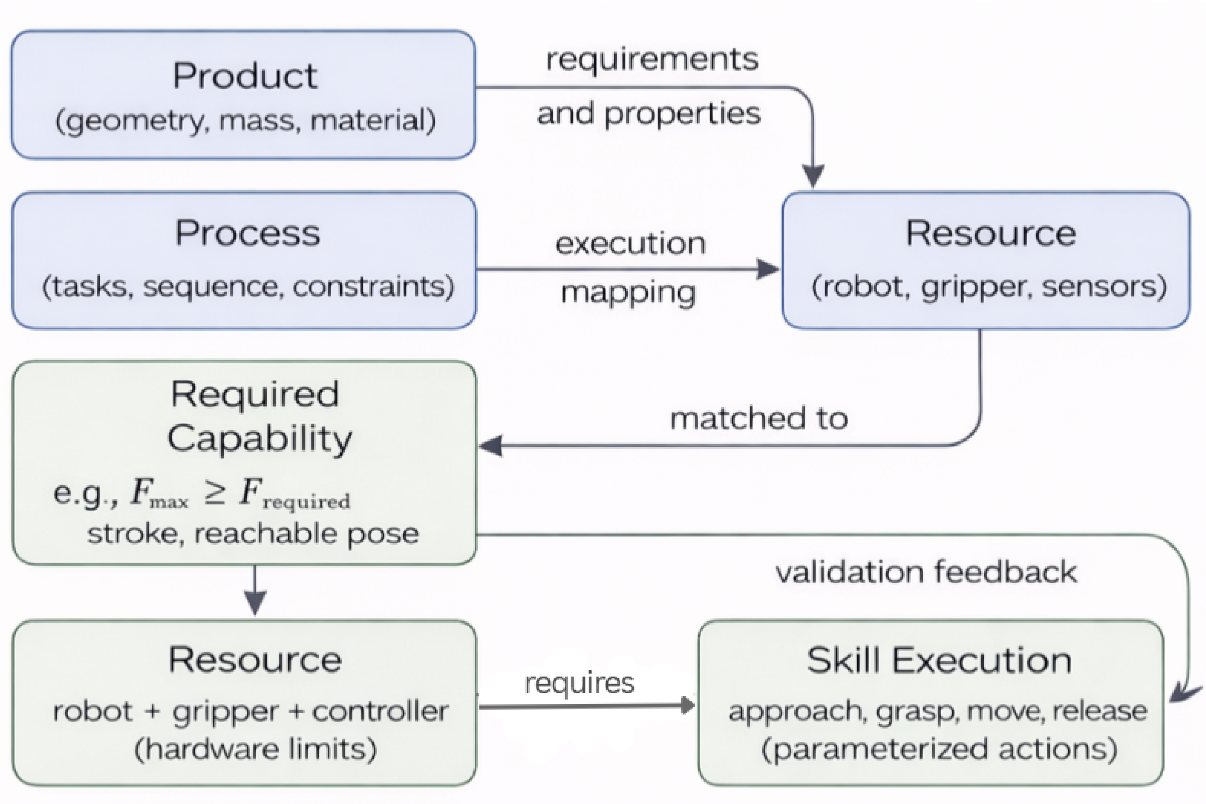

Fig. 2.2: Product-Process-Resource-Skill-Capability (PPR-SC) workflow.

Figure 2.2 illustrates the conceptual structure of the PPR-SC framework used in this thesis. In this model, the product defines the physical properties of the manipulated object, the process describes the required manipulation sequence, the capability layer defines the functional requirements needed to perform the task, and the resource layer represents the machines capable of fulfilling those requirements. The skill layer then represents the executable operations performed by the resource in order to complete the process.

Capability-based modelling has become an important concept in modern Industry 4.0 architectures, where production systems are increasingly designed as modular cyber-physical systems capable of dynamic reconfiguration. By describing production tasks in terms of capabilities rather than specific machines, automation systems can automatically identify suitable resources for executing a given process. Such approaches are particularly valuable in robotic automation systems where multiple robots or tools may be available to perform similar manipulation tasks. The use of capability-based modelling frameworks therefore enables more flexible task planning and improves the scalability of automated production systems [1], [23].

In the context of robotic manipulation, the PPR-SC framework provides a structured method for linking object properties, manipulation processes, and robotic capabilities. Within the framework proposed in this thesis, the product layer contains information extracted from the CAD model of the manipulated object. This information includes geometric features such as object dimensions, surface geometry, and the location of the center of mass. These parameters are essential for determining feasible gripping points and calculating the forces required for stable manipulation.

The process layer describes the sequence of actions required to execute the pick-and-place operation. Typical process steps include approaching the object, performing the grasping action, lifting the object, transporting it to a target location, and releasing it at the final position. Each of these actions corresponds to a specific manipulation step that must be executed by the robotic system. By representing the manipulation task as a structured sequence of process steps, the framework enables systematic planning and execution of robotic operations.

The resource layer represents the physical components used to execute the manipulation task. In this thesis, the primary resource is a robotic manipulator equipped with a parallel-jaw gripper. Additional resources include the simulation environment used for validating manipulation strategies and the software modules responsible for parameter estimation and motion control.

The capability layer defines the functional requirements needed to execute each step of the manipulation process. For example, stable grasping of an object requires the capability to generate sufficient gripping force to prevent slippage under gravitational and inertial loads. Similarly, the robot must possess the capability to reach the required spatial positions and orientations necessary for performing the pick-and-place task.

The skill layer represents the executable operations that implement these capabilities within the robotic control system. Skills correspond to parameterized robot actions such as approach, grasp, lift, transport, and release. These skills are executed by the robotic controller using the parameters generated through the analytical gripping parameter estimation process described later in this thesis.

Within the overall system architecture described in Section 3.2, the PPR-SC modelling framework serves as a knowledge representation layer connecting geometric data extraction, analytical parameter estimation, and simulation-based validation. The geometric information extracted from the CAD model is first used to compute gripping parameters such as gripping force, gripping location, and approach trajectory. These parameters are then stored in a structured data format and associated with specific manipulation skills within the PPR-SC framework. Finally, the robotic simulation environment executes these skills using the calculated parameters, enabling the verification of the generated manipulation strategy.

The integration of the PPR-SC modelling approach into the proposed workflow offers several advantages for robotic automation systems. First, it promotes modular system design by separating product information from process definitions and resource capabilities. Second, it enables greater flexibility by allowing tasks to be defined in terms of required capabilities rather than specific hardware implementations. Third, it facilitates interoperability between different engineering tools and automation platforms by providing a standardized representation of manufacturing knowledge.

Overall, the PPR-SC modelling framework provides a structured foundation for linking digital product models with robotic manipulation tasks. By integrating CAD-based product information, analytical parameter estimation, and capability-based resource modelling, the framework enables the automated generation and validation of gripping strategies for robotic pick-and-place operations. This modelling approach therefore plays a central role in the proposed methodology for automated gripping parameter estimation and forms a key component of the system architecture developed in this thesis.

3 Framework for Automated Gripping Parameter Estimation

3.1 Research Questions

The increasing demand for flexible manufacturing systems requires robotic manipulation processes that can rapidly adapt to variations in product geometry and assembly requirements. In industrial pick-and-place applications, particularly in modular battery assembly, gripping strategies are often manually configured by engineers through trial-and-error procedures in simulation environments. Such approaches are inefficient and difficult to scale in high-mix, low-volume production settings where product variants frequently change. As highlighted in the project motivation, modern production environments require automated and systematic approaches for deriving gripping parameters directly from digital product information.

To address these challenges, this thesis investigates the possibility of automatically determining gripping parameters from object geometry while integrating structured knowledge representation and simulation-based validation. The work focuses on establishing a methodology that links digital object models with robotic manipulation tasks through analytical models and knowledge-based reasoning. Based on this objective, the research presented in this thesis is guided by the following research questions.

RQ1 ***Which methodology can be used for gripping parameter estimation?***

RQ2 ***What data are necessary to calculate the required parameters for gripping and motion?***

RQ3 ***Can knowledge stored in an ontology be used to estimate gripping parameters, and which data must be represented to support automatic configuration of the resource model?***

RQ4 ***Can gripping parameters be calculated using simple mechanical formulas without relying on computationally intensive simulation software?***

RQ5 ***How do tolerances and elastic material properties affect parameter identification, and how can they be integrated into the data and methodology?***

Together, these research questions define the conceptual foundation of the proposed framework for automated gripping parameter estimation. They guide the development of the workflow architecture, analytical models, and knowledge representation mechanisms described in the following sections. By systematically addressing these questions, the thesis aims to contribute toward the development of more autonomous and adaptable robotic manipulation systems capable of configuring gripping strategies directly from digital product information.

3.2 System Architecture and Workflow Design

The objective of the proposed system architecture is to enable the automated estimation and validation of gripping parameters for robotic pick-and-place operations based on digital object models. The architecture integrates geometric data extraction, analytical parameter estimation, structured knowledge representation, and physics-based simulation into a unified workflow. By linking these components, the system provides a structured pipeline that transforms digital product information into executable robotic manipulation tasks. Such an integrated approach is particularly important in modern flexible manufacturing environments, where robotic systems must adapt to variations in product geometry and assembly requirements with minimal manual intervention.

In many industrial robotics applications, gripping parameters such as gripping force, grasp location, and approach trajectories are typically determined manually by engineers through trial-and-error procedures or repeated simulation experiments. While these approaches may work effectively in fixed production environments, they become inefficient in high-mix low-volume (HMLV) manufacturing scenarios where product geometries frequently change. Recent research in robotic manipulation has therefore focused on automated methods for grasp planning and parameter estimation based on geometric reasoning and analytical models [2], [5]. The system architecture proposed in this thesis follows this paradigm by combining CAD-based geometry extraction with simplified mechanical models to estimate gripping parameters in a systematic and reproducible manner.

The overall workflow of the proposed framework is illustrated in Fig 3.1. The architecture connects CAD-based geometry extraction, analytical gripping parameter estimation, knowledge-based modeling, and physics-based simulation validation in a unified pipeline.

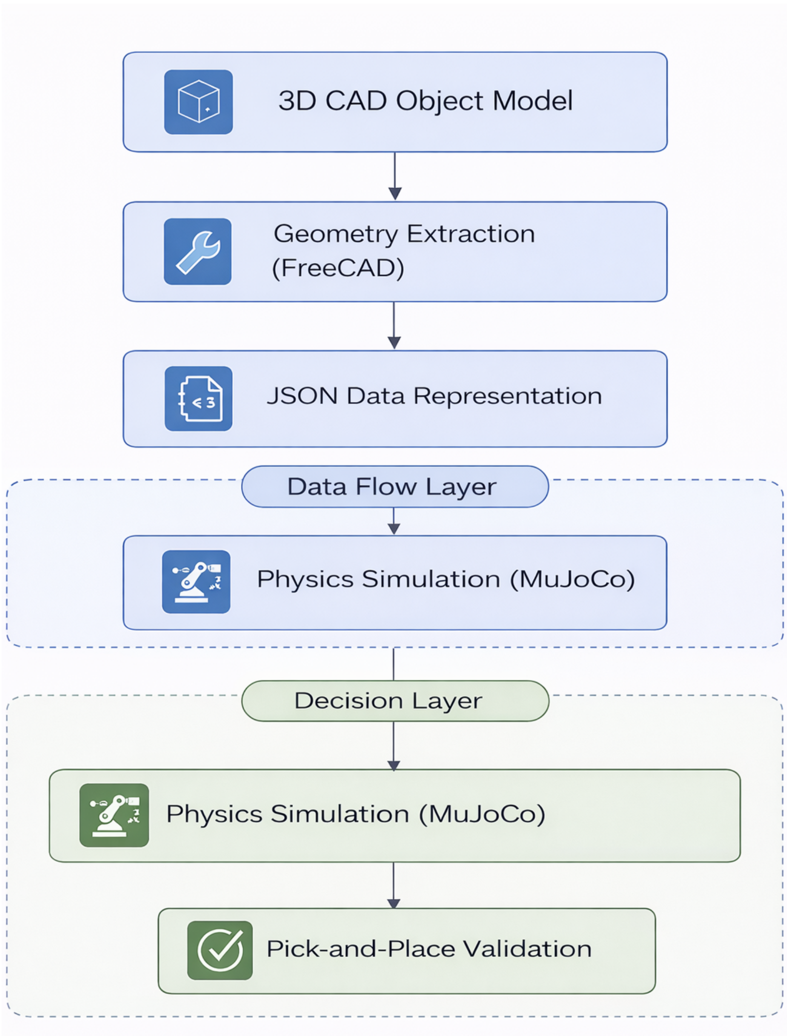

Fig. 3.1: System architecture and workflow for automated gripping parameter estimation.

The workflow begins with a digital representation of the manipulated object in the form of a three-dimensional CAD model. CAD models are widely used in manufacturing systems as digital descriptions of product geometry and design specifications. These models provide detailed information about the object's dimensions, shape, and spatial orientation, which can be used to determine feasible grasp locations and manipulation strategies. In addition to geometric information, CAD models may also contain physical properties such as mass distribution and the location of the center of gravity. Such information is essential for estimating the forces and torques involved in robotic grasping. Previous research has demonstrated that CAD-based grasp planning approaches can significantly improve the automation of robotic manipulation tasks by providing structured geometric input for grasp generation algorithms [4].

The next stage of the workflow focuses on extracting geometric features from the CAD model that are relevant for manipulation. In this work, the FreeCAD platform is used to access and process geometric data from the object model. FreeCAD provides programmable interfaces that allow automated extraction of object properties such as bounding box dimensions, surface geometry, and the position of the center of mass. These parameters serve as the primary inputs for the analytical gripping parameter estimation process. The extraction of geometric features from CAD models has been widely investigated in the context of automated assembly and grasp planning systems, where object geometry plays a central role in determining feasible grasp configurations [4], [10].

Once the geometric information has been extracted, the relevant parameters are stored in a structured data representation. In this work, a JSON-based format is used to encode the extracted object properties and the parameters required for gripping analysis. JSON provides a lightweight and flexible format that can easily be interpreted by different software components within the workflow. The JSON structure typically contains information such as object dimensions, center of mass location, potential gripping points, and estimated gripping forces. Using a structured data representation ensures interoperability between the geometry extraction stage and the subsequent decision-making components of the system architecture.

Following the data extraction stage, the workflow enters the analytical parameter estimation phase. In this phase, simplified mechanical models derived from classical mechanics are used to estimate the gripping parameters required for stable object manipulation. These calculations are primarily based on friction constraints and equilibrium conditions between the gravitational force acting on the object and the contact forces applied by the gripper. Analytical grasp models based on force closure and friction cone analysis have been extensively studied in robotic manipulation literature and form the basis for many grasp quality evaluation methods [5], [8]. By applying these principles, the system estimates the minimum gripping force required to prevent object slippage during manipulation.

In addition to estimating the gripping force, the analytical model also determines auxiliary parameters required for executing the pick-and-place operation. These include the optimal gripping location, the orientation of the gripper relative to the object, and the pre-grasp and post-grasp positions used during the manipulation sequence. These parameters define the spatial configuration of the robotic manipulator and ensure that the object can be grasped, transported, and released without collisions or loss of stability. Analytical methods for grasp pose generation have been successfully applied in industrial applications due to their computational efficiency and interpretability compared to purely data-driven approaches [10].

The estimated gripping parameters are then integrated into a knowledge-based representation based on the Product–Process–Resource Skill Capability (PPR-SC) model. The PPR-SC framework provides a structured method for representing manufacturing tasks by describing the relationships between the product being manipulated, the process steps required to perform the task, and the resources available to execute the operation. Within this framework, the product corresponds to the manipulated object, the process describes the sequence of actions involved in the pick-and-place task, and the resource represents the robotic manipulator and gripper used in the operation. Knowledge-based modeling approaches such as PPR-SC allow robotic systems to organize task information in a modular manner and enable more flexible automation architectures [1].

Finally, the estimated gripping parameters are validated using a physics-based simulation environment. In this thesis, the MuJoCo physics engine is used to simulate the interaction between the robotic manipulator, the gripper, and the manipulated object. MuJoCo provides accurate modeling of rigid-body dynamics and contact interactions, making it suitable for evaluating robotic manipulation tasks under realistic physical conditions [7]. Within the simulation environment, the robot and its surroundings are described using an XML-based model that defines the kinematic structure of the manipulator, the physical properties of the gripper, and the contact behavior between interacting bodies. A Python control script reads the gripping parameters stored in the JSON file and executes the corresponding pick-and-place operation within the simulation environment.

The simulation stage plays a crucial role in verifying the feasibility of the gripping strategy generated by the analytical model. While analytical calculations provide efficient methods for estimating gripping parameters, they often rely on simplified assumptions that may not fully capture complex contact dynamics or environmental interactions. Physics-based simulation allows these assumptions to be evaluated under more realistic conditions and provides valuable feedback for refining the estimated parameters. Similar simulation-based validation approaches have been widely used in robotic manipulation research to evaluate grasp stability and manipulation feasibility before deployment in real robotic systems [6], [15].

Overall, the proposed system architecture establishes a comprehensive pipeline that connects digital product models with robotic manipulation tasks. By integrating CAD-based geometry extraction, analytical gripping parameter estimation, structured knowledge representation, and physics-based simulation validation, the architecture enables the automated generation and verification of gripping strategies. This workflow provides a scalable approach for configuring robotic pick-and-place operations in flexible manufacturing environments and forms the foundation for the implementation and experimental evaluation presented in the subsequent chapters of this thesis.

3.3 Data Requirements for Gripping and Motion Parameterization

Robotic pick-and-place operations require the accurate configuration of several parameters that determine whether an object can be grasped, transported, and placed reliably within a manufacturing process. These parameters include the gripping location, gripping force, approach configuration, and motion trajectory of the robotic manipulator. The calculation of such parameters requires structured information about the manipulated object, the manipulation process, and the capabilities of the robotic system performing the task. In automated manufacturing environments, this information must be extracted from digital engineering data sources and organized in a structured format suitable for analytical computation and simulation-based validation.

In the proposed framework, gripping parameters are derived from a combination of geometric, physical, process-related, and resource-related data. These data categories correspond to the Product–Process–Resource modelling paradigm discussed in Section 2.5 and serve as the primary inputs for the analytical gripping parameter estimation method described in the following section. The accurate identification and representation of these parameters is therefore a crucial prerequisite for automated manipulation planning.

Previous research in robotic grasp planning has shown that grasp stability and manipulation feasibility depend strongly on object geometry, mass properties, contact conditions, and the physical capabilities of the robotic gripper [2], [5]. In model-based grasp planning approaches, these parameters are typically derived from digital object models or predefined engineering databases. CAD-based manipulation planning systems, for example, use geometric information from 3D object models to identify feasible grasp locations and evaluate stability conditions before executing the manipulation task [4], [10].

This section therefore analyzes the categories of data required for the automated estimation of gripping parameters and describes how these parameters are obtained from digital product models and system specifications.

3.3.1 Geometric Properties of the Manipulated Object

The geometric characteristics of the manipulated object represent the primary source of information required for determining feasible grasp configurations. In automated manipulation systems, these geometric properties are typically obtained from three-dimensional CAD models that provide a detailed representation of the object's shape, dimensions, and spatial orientation.

CAD models contain precise descriptions of the object geometry in the form of surfaces, edges, and volumetric representations. From these models, several geometric parameters relevant to robotic grasping can be extracted. These include:

- Object Dimension
- Bounding box
- Surface geometry
- Optimal Gripping location
- Orientation of the object

Bounding box dimensions are particularly useful for estimating approximate grasp locations for parallel-jaw grippers. The bounding box defines the minimum rectangular volume that contains the object geometry and provides a simplified representation of the object's size. This representation is frequently used in grasp planning algorithms to estimate feasible contact locations and gripper opening requirements [10].

Surface geometry also plays an important role in determining whether a stable grasp can be achieved. Flat surfaces are typically preferred for parallel-jaw gripping because they provide larger contact areas and more predictable friction conditions. Curved or irregular surfaces may reduce the effective contact area and increase the likelihood of slippage.

The geometric data extracted from CAD models therefore form the basis for determining potential grasp points and evaluating whether the object can be securely held by the gripper.

3.3.2 Physical Properties of the Object

In addition to geometric information, the physical properties of the object are essential for determining the forces required for stable manipulation. These properties influence the mechanical interaction between the gripper and the object and determine whether the applied gripping forces are sufficient to prevent slippage or rotational instability.

The most important physical parameters include:

- Object mass
- Center of mass location
- Object Material
- Surface friction

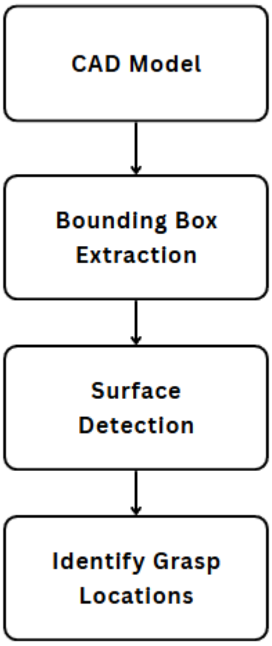

Fig. 3.2: CAD-based extraction of geometric parameters for gripping analysis.

The mass of the object directly affects the gravitational force acting on the object during lifting operations. Accurate knowledge of the mass is therefore required for estimating the forces that must be counteracted by the gripping mechanism.

The location of the center of mass is another critical parameter. If the gripping points are not aligned with the center of mass, gravitational forces generate a torque that tends to rotate the object within the gripper. This rotational effect must be considered during gripping parameter estimation to ensure stable manipulation.

Material properties and surface characteristics influence the coefficient of friction between the gripper fingers and the object surface. The friction coefficient determines the maximum tangential force that can be applied at the contact interface without causing slip, according to Coulomb's friction law [5].

Accurate estimation of friction coefficients is often challenging in practical applications because friction depends on several factors including surface roughness, material composition, and environmental conditions. In industrial practice, typical friction coefficients are often obtained from experimental measurements or engineering reference tables.

3.3.3 Gripper and Resource Parameters

While object properties determine the mechanical requirements for stable grasping, the capabilities of the robotic resource define whether these requirements can actually be ful-

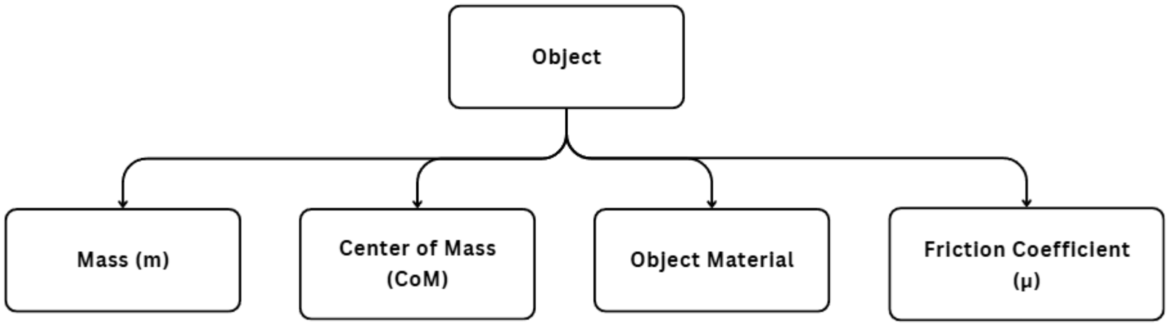

Fig. 3.3: Physical parameters of the manipulated object relevant for gripping stability analysis.

filled. The robotic manipulator and gripper must possess sufficient mechanical capabilities to generate the required forces and perform the necessary motion sequences.

Important resource parameters include:

- Maximum Gripping Force.
- Gripper Stroke.
- Finger Geometry.
- Robot reachability.
- Robot motion speed.

The maximum gripping force defines the upper limit of the force that can be applied by the gripper. If the analytically required gripping force exceeds this limit, the grasp configuration is infeasible.

The gripper stroke determines the maximum opening distance between the gripper fingers. This parameter must be large enough to accommodate the object dimensions while still allowing sufficient closing distance to apply the required gripping force.

Robot reachability is another important constraint. The robotic manipulator must be able to reach the calculated gripping position without violating joint limits or colliding with surrounding objects. Reachability analysis is therefore an essential part of robotic manipulation planning.

These resource parameters are typically obtained from the technical specifications of the robotic hardware or from the digital models used within the simulation environment.

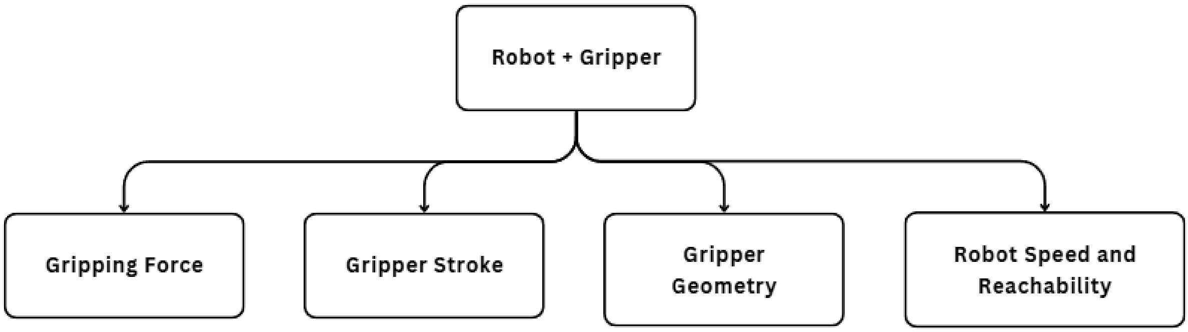

Fig. 3.4: Resource parameters influencing gripping feasibility.

3.3.4 Process Parameters for Pick and Place Operations

In addition to product and resource properties, manipulation planning also requires information about the process itself. Process parameters define how the object should be manipulated within the manufacturing workflow and specify constraints related to the pick-and-place task.

Typical process parameters include:

- Approach direction.
- Pre-Grasp position.
- Post-Grasp position.
- Placement Orientation.
- Release height.

The pre-grasp position represents the location from which the robot approaches the object before executing the grasp. This position must be chosen to avoid collisions and ensure that the gripper is properly aligned with the object.

Similarly, the post-grasp position defines the configuration of the robot after the object has been lifted. This position ensures that the object can be safely transported to the target location.

These process parameters are typically defined within the task planning layer of the automation system and must be compatible with both the object geometry and the robot's motion capabilities.

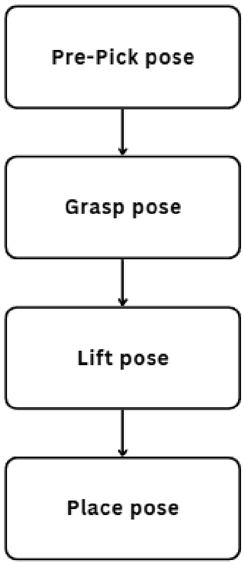

Fig. 3.5: Process parameters defining the pick-and-place motion sequence.

3.3.5 Structured Representation of Gripping Data

Once the relevant parameters have been identified, they must be organized in a structured format that allows efficient processing by the analytical models and simulation environment. In modern digital manufacturing systems, structured data representation plays an important role in enabling interoperability between different engineering tools and software platforms.

The proposed framework organizes the required parameters according to the Product–Process–Resource modelling paradigm introduced earlier. In this representation:

- Product data include geometric and physical properties of the object.
- Process data describe the manipulation sequence and motion parameters.
- Resource data define the capabilities of the robotic manipulator and gripper.

This structured representation ensures that all relevant parameters required for gripping analysis are available in a consistent format and can be easily accessed by different components of the system architecture.

The parameters identified in this section represent the essential input data required for the analytical estimation of gripping forces and manipulation configurations. However, the existence of these parameters alone does not define how they should be used to compute the final gripping strategy. A systematic methodology is therefore required to transform the extracted geometric and physical information into executable robotic manipulation parameters.

The following section therefore introduces the analytical methodology used in this thesis to derive gripping forces, grasp locations, and motion parameters based on the data categories described above.

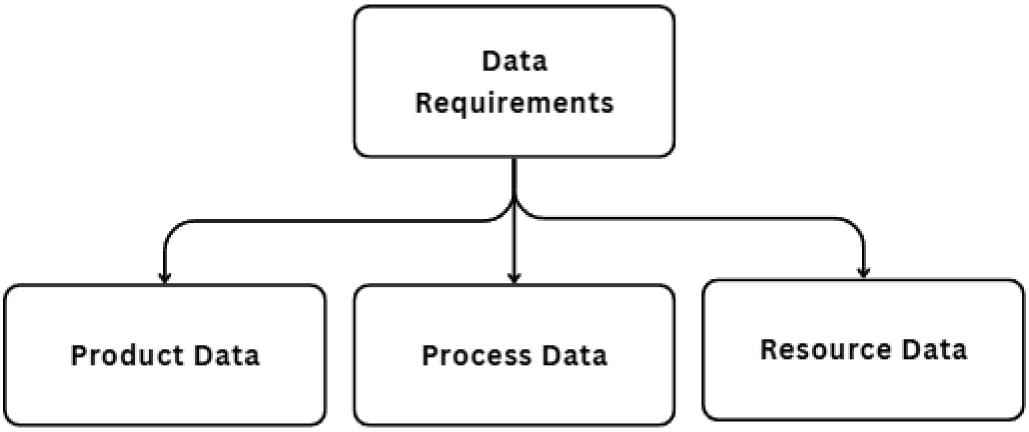

Fig. 3.6: Data categories required for gripping parameter estimation within the proposed framework.

4 Analytical Modelling and Knowledge Integration

4.1 Analytical Derivation of Gripping Parameters

Robotic grasp planning involves determining the parameters required for a manipulator to securely grasp and manipulate an object without causing slippage or instability. In industrial pick-and-place applications, these parameters typically include the gripping location, gripping force, approach direction, and motion trajectory. While modern robotic systems increasingly incorporate data-driven or learning-based methods for grasp generation, analytical models derived from classical mechanics remain widely used in industrial environments due to their transparency, computational efficiency, and ease of implementation [2], [5]. Analytical methods provide a systematic approach for estimating the forces and constraints involved in stable object manipulation, allowing gripping strategies to be derived directly from geometric and physical object properties.

In the framework proposed in this thesis, gripping parameters are calculated using analytical models based on rigid-body mechanics and frictional contact theory. These calculations rely on geometric information extracted from the CAD model of the object as well as physical parameters such as mass and friction coefficients. The resulting parameters are then used to generate the input data required for the simulation-based validation stage described later in this thesis.

4.1.1 Grasp Stability and Contact Mechanics

The stability of a robotic grasp depends on the ability of the gripper to apply sufficient forces at the contact points to counteract external disturbances such as gravity and inertial loads. In the case of a parallel-jaw gripper, the object is typically held between two opposing contact surfaces. The stability of this grasp can be analyzed using friction-based contact models, where the tangential forces at the contact interface must remain within the limits defined by Coulomb's friction law.

The classical friction constraint can be expressed as in Equation 2.4

$$|F_t| \leq \mu F_n$$

where $\boldsymbol{F_t}$ is Tangential Friction Force, $\boldsymbol{F_n}$ is the normal contact force and $\boldsymbol{\mu}$ is the friction coefficient between the gripper and the finger.

This inequality defines the friction cone constraint, which describes the set of allowable contact forces that prevent slippage at the contact interface [5], [24]. When the tangential force exceeds this limit, the object begins to slide relative to the gripper surface, leading to grasp failure.

The gravitational force $\boldsymbol{F_g}$ acting on the object is given by Equation 2.3

$$\mathrm{F}_g = mg$$

where **m** is the object mass and **g** is the gravitational acceleration.

In a symmetric two-finger grasp, the gravitational force must be balanced by the friction forces generated at the two contact points. Assuming equal force distribution between the fingers, the minimum gripping force required to prevent slipping can be derived to Equation 2.10

$$F_n \geq \frac{mg}{2\mu}$$

This equation provides a simple analytical estimate for the minimum normal force required to maintain a stable grasp under static conditions.

4.1.2 Torque and Moment Equilibrium

While the friction constraint ensures that the object does not slide out of the gripper, rotational stability must also be considered. If the center of mass of the object is not aligned with the gripping points, gravitational forces may create a torque that tends to rotate the object within the gripper.

The torque generated by gravitational forces can be expressed as in Equation 2.12

$$M_g = mg(d + \Delta d)$$

where **d** is the nominal distance between the gripping point and the center of mass and $\boldsymbol{\Delta d}$ is the positional tolerance or uncertainity.

This torque must be balanced by the moment generated by the gripping forces applied by the gripper fingers. Assuming a contact distance **r** from the center of rotation, the resisting torque generated by the gripping forces is:

$$M_r = 2\mu F_n r \tag{4.1}$$

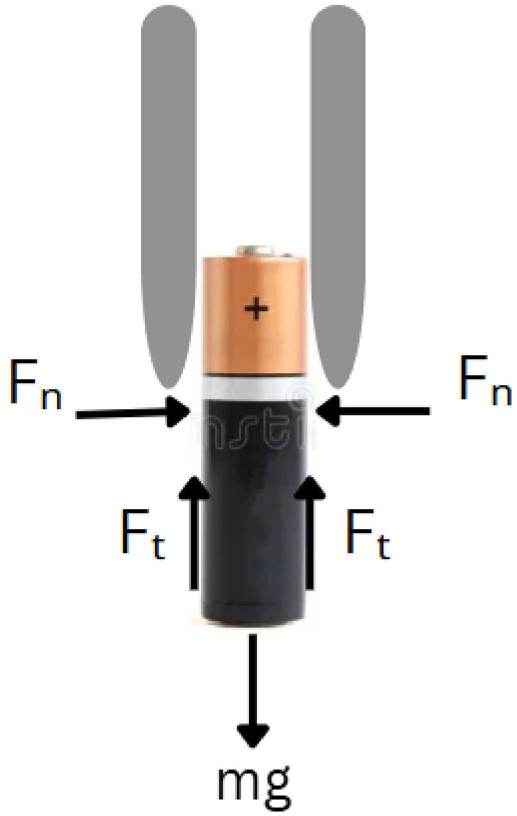

Fig. 4.1: Friction-based force model for a two-finger parallel gripper grasp illustrating the Coulomb friction constraint.

To ensure rotational stability, the resisting torque must exceed the gravitational torque:

$$2\mu F_n r \geq mg(d + \Delta d) \tag{4.2}$$

Solving for the normal gripping force yields:

$$F_n \geq \frac{mg(d + \Delta d)}{2\mu r} \tag{4.3}$$

This equation shows that the required gripping force increases when the center of mass is further away from the gripping point or when the friction coefficient decreases.

4.1.3 Combined Gripping Force Requirement

In practice, both translational and rotational stability constraints must be satisfied simultaneously. Therefore, the required gripping force must be chosen as the maximum value derived from both conditions. The combined force requirement can therefore be written as:

$$F_{required} \geq \max\left(\frac{mg}{2\mu}, \frac{mg(d + \Delta d)}{2\mu r}\right) \tag{4.4}$$

This equation forms the core analytical model used in this thesis for estimating the gripping force required for stable manipulation. The parameters used in the equation are obtained directly from the CAD model of the object and the material properties of the contact surfaces.

4.1.4 Dynamic Considerations

Although the previous analysis assumes static conditions, robotic pick-and-place operations often involve dynamic motion during object lifting and transportation. During these motions, additional inertial forces may act on the object. According to Newton's second law, the dynamic motion of the object can be described by Equation 2.16

$$m\ddot{x} = \sum F \tag{4.5}$$

Similarly, rotational dynamics can be expressed by Equation 2.17

$$I\dot{\omega} = \sum M \tag{4.6}$$

where $\mathbf{I}$ is the moment of inertia and $\boldsymbol{\omega}$ is the angular velocity.

These equations describe the relationship between applied forces, torques, and the resulting motion of the object [3]. In practical applications, safety factors are often introduced to ensure that the gripping force remains sufficient even under dynamic conditions.

4.1.5 Extraction of Parameters from CAD Models

In the proposed framework, the parameters required for the analytical calculations are extracted directly from the CAD model of the object. Using the FreeCAD environment, geometric properties such as object dimensions, bounding box size, and the location of the center of mass can be computed automatically. These parameters form the basis for calculating the gripping force and determining suitable gripping locations.

CAD-based approaches to grasp planning have been widely studied in robotic manipulation research. By analyzing object geometry and surface features, it is possible to identify feasible gripping regions and evaluate grasp stability before executing the manipulation task. Such methods provide an efficient alternative to purely sensor-based grasp detection approaches and enable the automation of manipulation parameter generation directly from digital product models [4].

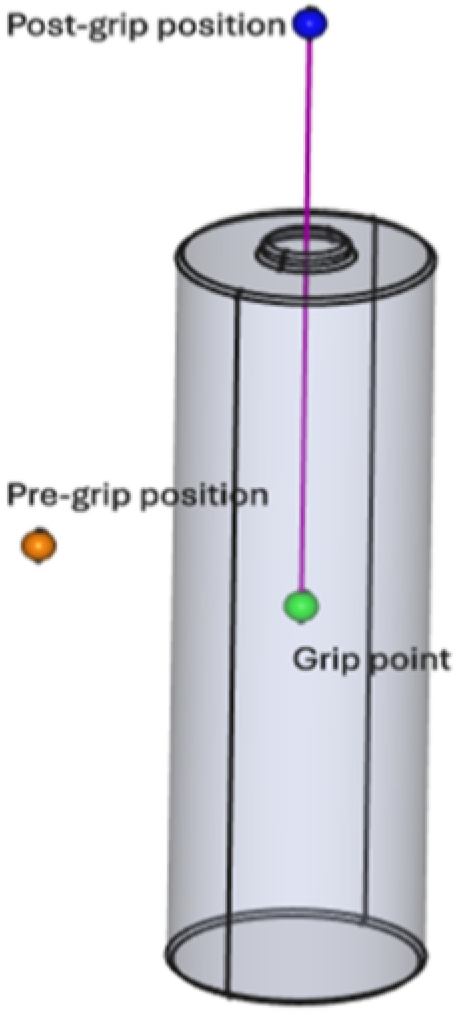

Fig. 4.2: Example of extracted parameter from CAD model

4.1.6 Integration with the System Workflow

The analytical gripping parameters calculated in this section form an essential part of the overall workflow architecture described in Section 3.2. After extracting geometric parameters from the CAD model, the gripping force and gripping location are calculated using the analytical models presented above. These parameters are then stored in a structured JSON format that serves as the interface between the parameter estimation stage and the robotic simulation environment.

These parameters are subsequently read by the Python control script used to execute the robotic manipulation task within the MuJoCo simulation environment. The simulation stage allows the calculated parameters to be validated under realistic physical conditions, including contact forces and friction interactions [7].

5 Implementation and Evaluation

5.1 CAD-Based Geometry Processing and Parameter Extraction (FreeCAD)

5.1.1 Role of CAD models in the Framework

In automated robotic manipulation systems, accurate knowledge of object geometry is essential for determining feasible grasp configurations and motion trajectories. Traditionally, grasp planning approaches rely on sensor-based perception methods such as vision systems or depth cameras to estimate object pose and geometry in real time. While such methods provide flexibility in unstructured environments, they can introduce uncertainty due to sensor noise, occlusions, and limited resolution.

In contrast, many industrial automation scenarios operate with well-defined products whose geometric models are already available in digital form.CAD models provide detailed information about the object's geometry, dimensions, and spatial features. These models can therefore serve as a reliable source of information for robotic manipulation planning. By analyzing CAD models directly, it becomes possible to extract relevant parameters for grasp planning and motion planning without relying solely on sensor-based detection methods.

CAD-based grasp planning has been widely studied in robotic manipulation research. Approaches based on geometric analysis of digital object models can identify candidate gripping regions, evaluate grasp stability conditions, and estimate manipulation forces before executing the manipulation task [4], [10]. Such methods enable the automation of parameter generation directly from digital product data and are particularly well suited for industrial production environments where object models are available during system design.

In the framework developed in this thesis, CAD models represent the starting point of the parameter estimation pipeline. The geometric properties extracted from the CAD model serve as inputs to the analytical gripping model described in Chapter 4 and enable the automated determination of gripping locations and gripping forces. This approach supports the overall goal of generating robotic manipulation parameters directly from digital engineering data.

5.1.2 FreeCAD Environment for Geometry Processing

To implement the CAD-based parameter extraction process, the open-source CAD platform FreeCAD was used. FreeCAD provides a flexible environment for geometric modelling and supports the automated extraction of geometric properties through its integrated Python scripting interface.

FreeCAD offers several advantages for the proposed framework:

- Support for common CAD file formats such as STEP and STL.
- Built-in tools for calculating geometric and physical properties.
- Python-based scripting capabilities for automation.
- Compatibility with external robotics software environment

The Python interface of FreeCAD enables the automated processing of CAD models and the extraction of relevant parameters without manual interaction. This capability is particularly important for integrating CAD analysis into automated parameter estimation workflows.

Within the proposed system architecture, FreeCAD is used to compute several geometric properties of the manipulated object, including the object's dimensions, bounding box size, and center of mass location. These parameters are then exported for further processing by the analytical gripping model and simulation environment.

5.1.3 Extraction of Geometric Parameters

The geometric properties required for gripping parameter estimation are extracted directly from the CAD model using FreeCAD's geometric analysis tools. These properties describe the physical shape and spatial characteristics of the manipulated object and serve as inputs for determining feasible grasp configurations.

The following geometric parameters are extracted:

- Object dimensions.
- Bounding box dimensions.

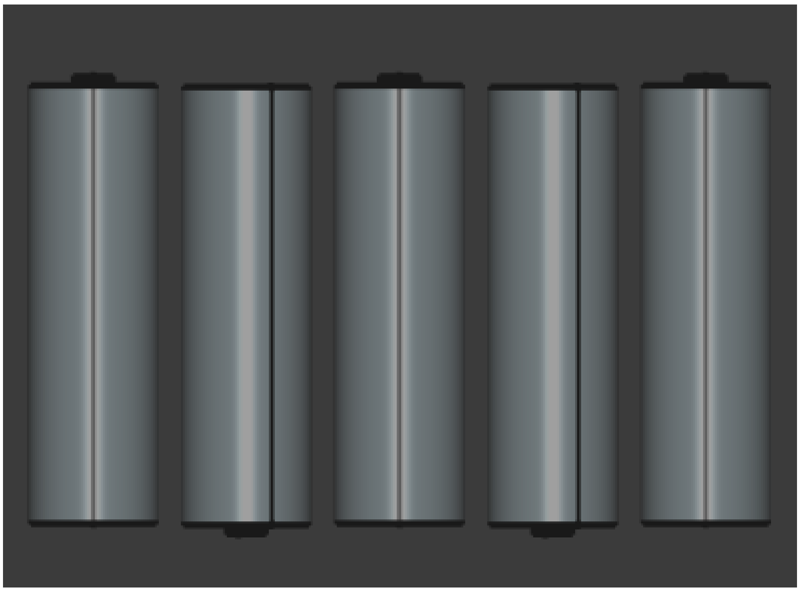

Fig. 5.1: CAD model of the manipulated object used for geometric parameter extraction.

- Surface geometry and potential contact area.
- Center of Mass (COM) position.

The extraction process is done by the code shown in the Listing 5.1

Listing 5.1: Geometry Data Extraction from FreeCAD Object

```
# --- GEOMETRY DATA EXTRACTION ---
if hasattr(obj, "Mesh"):
    mesh = obj.Mesh
    bb = mesh.BoundBox
    com = bb.Center
    volume_mm3 = mesh.Volume

elif hasattr(obj, "Shape"):
    shape = obj.Shape
    bb = shape.BoundBox
    com = shape.CenterOfMass
    volume_mm3 = shape.Volume

else:
    App.Console.PrintError("Object type not supported.\n")
    return
```

The results of the extraction are shown below in the table below

Tab. 5.1: Example of extracted geometric parameters

Parameter	Value
Object Length	14 mm
Object Width	82 mm
Object Height	51 mm
Center of Mass	(0, 0, 0 mm)
Bounding Box	(14 × 82 × 51 mm)

5.1.4 Automated Parameter Extraction Workflow

To enable automated processing of CAD models, a Python-based script was developed within the FreeCAD environment. This script loads the CAD model, computes the required geometric parameters, and exports the results in a structured format suitable for further processing.

The automated parameter extraction workflow consists of the following steps:

- Import CAD model into FreeCAD environment.
- Compute the bounding box of the object geometry.
- Extract relevant parameters.
- Export those parameters for further process.

This automated workflow ensures that geometric information can be obtained directly from the digital product model without manual intervention.

The geometric parameters extracted from the CAD model represent essential inputs for the gripping parameter estimation process. However, these parameters must be organized in a structured format that allows them to be accessed by the robotic simulation and control system.

To achieve this, the extracted parameters are stored in a structured data representation that enables communication between the parameter extraction module and the robotic simulation environment. The following section therefore describes the JSON-based data interface used to represent gripping parameters and manipulation configurations within the proposed framework.

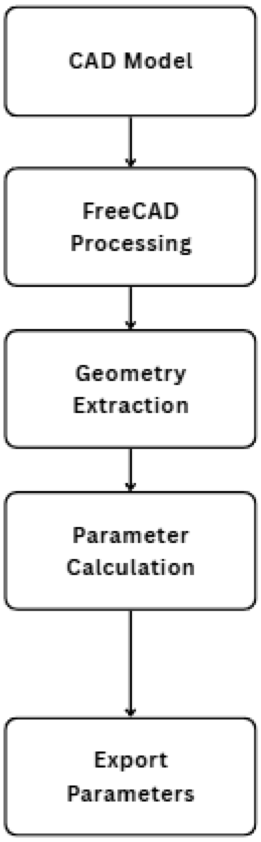

Fig. 5.2: Automated workflow for extracting geometric parameters from CAD models.

5.2 JSON-Based Data Modelling and Interface Design

The analytical gripping parameters derived in the previous section must be transferred to the robotic execution system in a structured and machine-readable format. In modern robotic automation systems, the use of standardized data representations plays an essential role in enabling interoperability between different software components. In the framework developed in this thesis, the calculated gripping parameters are stored in a JSON structure that serves as the interface between the parameter generation stage and the simulation environment.

JSON is a lightweight and widely used data-interchange format that allows structured information to be represented using key–value pairs and hierarchical objects. Due to its simplicity and compatibility with many programming languages, JSON has become a common format for exchanging structured data between software systems in robotics, automation, and web-based applications. The use of JSON also enables easy integration with Python-based simulation scripts and external software tools, which makes it particularly suitable for the proposed workflow.

5.2.1 Parameter mapping from Analytical Model

The gripping parameters computed using the analytical model described in Section 3.3 must be mapped into a structured representation that can be interpreted by the robotic control software. These parameters include both object-specific properties and task-related parameters required for executing the pick-and-place operation.

The primary parameters stored in the JSON structure include:

- Object mass - Given Manually
- Center of mass position - From the FreeCAD Macro (fixed at origin in this case)
- Gripping point
- Required gripping force
- Pre-grasp position
- Post-grasp position
- Gripper open width

The gripping force is computed directly within the FreeCAD macro using the extracted object volume and material density. The required normal force per jaw is calculated as shown in equation 5.1, which extends the formula 2.6 by introducing a safety factor.

$$F_{per-jaw} = \frac{(m * g * S_f)}{2\mu} \tag{5.1}$$

Here the safety factor $\boldsymbol{S_f}$ and the friction coefficient $\boldsymbol{\mu}$ depends on the object and gripper material. This calculation is shown in Listing 5.2

Listing 5.2: Gripping Strength

```
# ==========================
# CONFIGURATION
# ==========================
DENSITY_KG_PER_M3 = 2700   # Aluminium
MU = 0.5
SAFETY = 2.0
G = 9.80665
```

```
 # Mass & Force
    volume_m3 = abs(volume_mm3) * 1e-9
    mass = volume_m3 * DENSITY_KG_PER_M3
    f_per_jaw = (mass * G * SAFETY) / (2 * MU)

"force_estimation":
    {
        "mass_kg": r(mass),
        "friction_coefficient": MU,
        "safety_factor": SAFETY,
        "normal_force_per_jaw_N": r(f_per_jaw),
        "total_normal_force_both_jaws_N": r(2 * f_per_jaw)
    },
```

The Grip positions are calculated from the object's bounding box, top surface and the center of mass. The Python Macro first gets the bounding box **bb**, center of mass **COM**, and volume from the selected mesh. It takes the object's top surface as $z_{max} = bb.ZMax$.

The grip point is calculated near the top of the object. The Pre-Grasp point is calculated simply by placing it vertically above the object, at an offset above the top surface which is arrived by means of Iterations of several values within a certain range. So the pre-grasp is defined only as a Z offset relative to COM, with X and Y set to zero. The Post-Grasp point calculation also follows the same method but in the opposite direction. The code snippet is given below at Listing 5.3

Listing 5.3: Gripping Point Positions

```
# --- BASIC GEOMETRY ---
z_max = bb.ZMax
x_c = 0.5 * (bb.XMin + bb.XMax)
y_c = 0.5 * (bb.YMin + bb.YMax)

# --- AUTOMATIC SEARCH LIMIT ---
bottom_limit_from_top_mm = clamp(
    height * BOTTOM_LIMIT_FROM_TOP_RATIO,
    MIN_BOTTOM_LIMIT_FROM_TOP_MM,
    MAX_BOTTOM_LIMIT_FROM_TOP_MM
)

# Generate candidate grip depths from the top surface
   downward
grip_depth_values = linspace(0.0, bottom_limit_from_top_mm,
   NUM_GRIP_LEVELS)
```

```
# --- GRIP POINT CANDIDATE GENERATION ---
for grip_depth in grip_depth_values:
    # Initial grip-point guess at XY center and selected
       depth below top surface
    grip_guess = App.Vector(x_c, y_c, z_max - grip_depth)

    # Project the guessed point onto the actual object
       surface
    grip_world = grip_guess
    if shape is not None:
        try:
            dist, pts, info = shape.distToShape(Part.Vertex(
               grip_guess))
            if pts[0] and hasattr(pts[0][0], "x"):
                grip_world = pts[0][0]
        except Exception:
            grip_world = grip_guess

    # Store grip point relative to the object's center of
       mass
    grip_rel = grip_world - com
```

The gripper opening is determined directly from the object geometry. Based on the top to down grasping approach, the relevant dimension is a smaller horizontal extent of the object, obtained from the bounding box. A similar approach is used to generate clearances through iterations and it is added to ensure collision-free approach (Listing 5.4). The resulting total opening width is then divided equally between both gripper fingers and stored in the JSON file for use in the simulation stage.

Listing 5.4: Gripper Open Width

```
# --- ESTIMATE OBJECT WIDTH FROM BOUNDING BOX ---
# Use the smaller dimension in XY as the gripping width
width = min(bb.XLength, bb.YLength)

# --- AUTOMATIC CLEARANCE CALCULATION ---
# Clearance values are derived as a fraction of object width
clearance_values = [
    clamp(width * factor, MIN_CLEARANCE_MM, MAX_CLEARANCE_MM)
    for factor in CLEARANCE_FACTORS
]
```

```
# Remove duplicates and round
clearance_values = sorted(set([r(c) for c in clearance_values
   ]))

# --- GRIPPER OPEN WIDTH CALCULATION ---
for clearance in clearance_values:
    # Gripper opening = object width + clearance on both
       sides
    gripper_open_width = width + 2.0 * clearance
```

This generates a series of grasp candidates and are serialized into a JSON file that acts as a configuration input for the simulation environment.

5.2.2 JSON Data Structure

The structure of the JSON file is designed to reflect the modelling approach described in the previous sections, where product properties, manipulation parameters, and robot control information are clearly separated.

A simplified example of the JSON structure used in this thesis is shown below.

Listing 5.5: Snippet of JSON File

```
    "object_id": "battery_grip",
    "relative_to_tcp": [
        0.0,
        0.0,
        0.0
    ],
    "source": "freecad_grip_macro
    "geometry": {
        "bounding_box_mm": [
            14.499348,
            82.498596,
            51.0
        ],
        "center_mm": [
            0.0,
            0.0,
            0.0
        ],
        "principal_axis_world": {
            "axis_dir_unit": [
                0.0,
```

```
                0.0,
                1.0
            ]
        }
    },
    "force_estimation": {
        "mass_kg": 0.109094,
        "friction_coefficient": 0.5,
        "safety_factor": 2.0,
        "normal_force_per_jaw_N": 2.139686,
        "total_normal_force_both_jaws_N": 4.279371
    },
    "candidates": [
        {
            "candidate_id": 1,
            "grip_depth_below_top_mm": 0.0,
            "grip_point_mm": [
                0.0,
                0.0,
                25.5
            ],
            "pre_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "post_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "clearance_mm": 0.43498,
            "estimated_object_width_mm": 14.499348,
            "gripper_open_width_mm": 15.369308,
            "pre_grip_offset_mm": 10.2,
            "release_height_mm": 20.0
        },
        {
            "candidate_id": 2,
            "grip_depth_below_top_mm": 0.0,
            "grip_point_mm": [
                0.0,
                0.0,
                25.5
            ],
```

```
            "pre_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "post_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "clearance_mm": 0.724967,
            "estimated_object_width_mm": 14.499348,
            "gripper_open_width_mm": 15.949282,
            "pre_grip_offset_mm": 10.2,
            "release_height_mm": 20.0
        },
        {
            "candidate_id": 3,
            "grip_depth_below_top_mm": 0.0,
            "grip_point_mm": [
                0.0,
                0.0,
                25.5
            ],
            "pre_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "post_grip_mm": [
                0.0,
                0.0,
                35.7
            ],
            "clearance_mm": 1.159948,
            "estimated_object_width_mm": 14.499348,
            "gripper_open_width_mm": 16.819244,
            "pre_grip_offset_mm": 10.2,
            "release_height_mm": 20.0
        },
```


This structure provides a clear and modular representation of the parameters required for executing the robotic manipulation task. Each section of the JSON file corresponds to a specific category of information within the workflow:

- The object section describes physical properties extracted from the CAD model.
- The grasp section stores the gripping parameters calculated using the analytical model.

Using this structured representation allows the parameter estimation module and the robotic control system to operate independently while maintaining a standardized communication interface.

5.2.3 Integration with the Simulation Environment

The JSON file generated during the parameter estimation stage is subsequently used as input for the robotic simulation environment. In the implementation developed in this thesis, a Python-based control script reads the JSON file and extracts the required parameters to configure the robotic manipulation task within the MuJoCo simulation environment.

MuJoCo is a physics-based simulation engine designed for modelling rigid-body dynamics and contact interactions in robotic systems. It provides accurate simulation of physical phenomena such as friction, collisions, and joint dynamics, making it suitable for validating robotic manipulation strategies before deployment in real-world systems [7].

The simulation workflow follows the steps below:

1. Load the JSON parameter file.
2. Extract object properties and gripping parameters.
3. Configure the robot motion trajectory.
4. Execute the pick-and-place task in the simulation environment.

This architecture ensures that the analytical parameter estimation process remains independent from the simulation system while still allowing seamless integration between the two components.

5.2.4 Data Flow within the System Architecture

The role of the JSON interface within the overall workflow architecture is illustrated in Figure 5.3

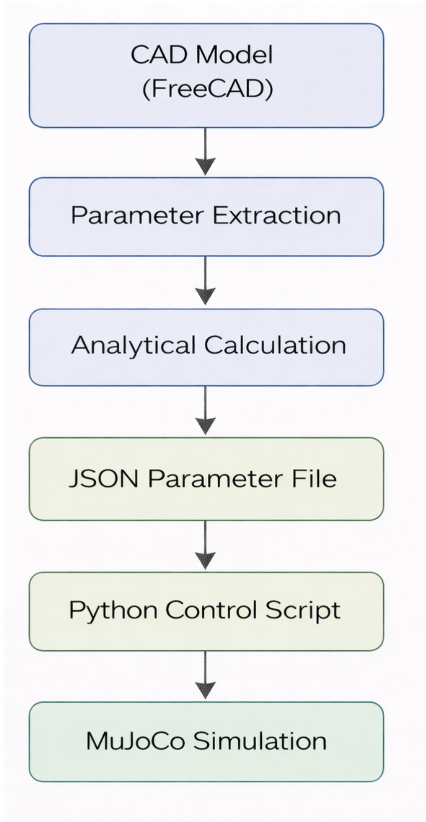

Fig. 5.3: An example of JSON file with the paramerters.

This data flow highlights how the proposed system integrates different software tools into a unified pipeline for generating and validating robotic manipulation parameters. The use of a structured data interface provides several advantages for robotic manipulation systems. First, it decouples the parameter estimation process from the simulation environment, allowing each component of the system to be developed and modified independently. Second, it enables easy integration with additional software modules, such as task planning algorithms or digital twin environments. Finally, it supports the reuse of computed parameters across multiple experiments or simulation scenarios.

Such modular data representations are increasingly used in modern robotic systems, where software components must exchange information across heterogeneous platforms and programming environments [1]. By adopting a standardized JSON interface, the framework developed in this thesis ensures that the generated gripping parameters can be easily integrated with both simulation environments and real robotic control systems.

5.3 Robot Modelling and Physics Simulation in MuJoCo

5.3.1 MuJoCo Simulation Environment

To validate the gripping parameters derived from the analytical model, a physics-based simulation environment was used. Simulation plays an important role in robotic manipulation research because it allows the evaluation of manipulation strategies under realistic physical conditions without the need for immediate deployment on physical hardware. By incorporating accurate models of rigid-body dynamics, contact forces, and friction interactions, simulation environments provide an effective platform for testing grasp stability and motion execution.

In this work, the **MuJoCo (Multi-Joint dynamics with Contact)** physics engine was used to simulate the robotic pick-and-place operation. MuJoCo is a widely used simulation framework designed for modelling articulated mechanical systems and their interaction with the environment. The simulator provides efficient numerical methods for computing rigid-body dynamics and contact interactions, making it particularly suitable for robotic manipulation tasks that involve complex contact dynamics between the robot gripper and manipulated objects.

MuJoCo offers several advantages for robotic manipulation simulation. First, it provides accurate modelling of contact forces and friction interactions, which are essential for evaluating grasp stability. Second, the simulator allows efficient computation of dynamic systems with multiple joints, enabling real-time simulation of complex robotic systems. Third, MuJoCo integrates well with Python-based control frameworks, allowing flexible development of custom control algorithms and parameter interfaces.

The physics engine implements a constraint-based formulation of rigid-body dynamics and contact interactions, enabling stable simulation of mechanical systems with multiple degrees of freedom. These capabilities have made MuJoCo a widely adopted tool in robotics research and control applications [7]. In the context of this thesis, the simulator is used to evaluate whether the gripping forces and grasp configurations derived from the analytical model can successfully perform the intended manipulation task.

5.3.2 Robot Model Representation

Within the MuJoCo simulation environment, robotic systems are described using XML-based model files. These XML files define the structure of the mechanical system, including rigid bodies, joints, actuators, and contact geometries. Each robot link is represented as a rigid body connected to other bodies through joints, forming a kinematic chain that describes the robot's motion capabilities.

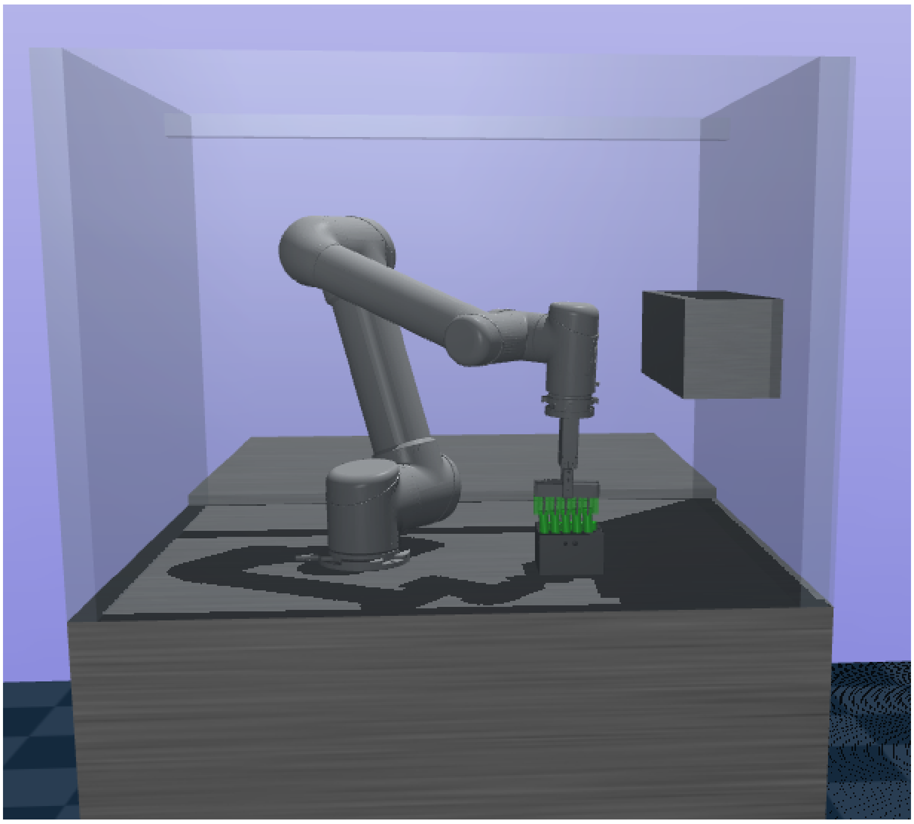

Fig. 5.4: MuJoCo simulation environment used to validate the robotic pick-and-place operation.

The robot model used in this thesis consists of multiple rigid bodies connected by revolute joints representing the robot's articulated structure. Each joint defines the allowed motion between two connected links and includes parameters such as joint limits, damping, and actuator properties. The kinematic structure of the robot therefore determines the reachable workspace and motion capabilities of the manipulator.

In addition to defining the kinematic structure, the XML model also specifies the geometric representation of each robot link. These geometries are used by the physics engine to detect collisions and compute contact forces during simulation. Accurate modelling of these geometries is essential for realistic simulation of robotic manipulation tasks, particularly when interactions between the gripper and the manipulated object are involved.

The robot model therefore provides the structural foundation of the simulation environment, enabling the execution of motion trajectories and the evaluation of manipulation strategies within a physically realistic context.

```xml
<mujoco model="ur5gripper">
    <worldbody>
        <!--Robot UR5-
        <body name="box_link" pos="0.2888 0.3355 0.93"> -->
        <body name="robot_cell" pos="0 0 0">

            <!-- Tischplatte -->
            <geom name="table_block" type="box" size="0.525 0.40 0.465" pos="0.525 0.40 0.465" contype="1" conaffinity="1" material="bench_mat" />
            <geom name="table_block_2" type="box" size="0.525 0.40 0.475" pos="0.525 1.20 0.475" contype="1" conaffinity="1" material="bench_mat" />
            <!-- Wand links (x = -0.02 + 0.02) = 0.02 -->
            <geom name="wall_left" type="box" size="0.02 0.40 0.37" pos="0.02 0.40 1.3" contype="1" conaffinity="1" material="glass"/>
            <!-- Wand rechts (x = 1.042 - 0.02 = 1.022) -->
            <geom name="wall_right" type="box" size="0.02 0.40 0.37" pos="1.022 0.40 1.3" contype="1" conaffinity="1" material="glass"/>
            <!-- Wand vorne (y = -0.02 + 0.02 = 0.02) -->
            <geom name="wall_front" type="box" size="0.525 0.02 0.37" pos="0.525 0.02 1.3" contype="1" conaffinity="1" material="glass"/>
            <!-- Decke -->
            <geom name="ceiling" type="box" size="0.525 0.02 0.02" pos="0.525 0.772 1.65" contype="1" conaffinity="1" material="glass"/>
            <!--GreiferFach-->
            <geom name="gripper_shelf" type="box" size="0.075 0.215 0.075" pos="0.945 0.425 1.275" contype="1" conaffinity="1" material="bench_mat"/>

            <inertial pos="0 0 0" mass="1000" diaginertia="0 0 0" />
            <body name="base_link" pos="0.3755 0.325 0.93" quat="-1 0 0 1" >
                <inertial pos="0 0 0" quat="0.5 0.5 -0.5 0.5" mass="4" diaginertia="0.0072 0.00443333 0.00443333" />
                <geom type="mesh"  mesh="base" material="ur5_mat"/>

                <body name="shoulder_link" pos="0 0 0.089159" >
                    <inertial pos="0 0 0" mass="3.7" diaginertia="0.0102675 0.0102675 0.00666" />
                    <joint name="shoulder_pan_joint" class="UR5" pos="0 0 0" axis="0 0 1" limited="true" range="-6.28318 6.28318" />
                    <geom name="shoulder_geom" type="mesh"  mesh="shoulder" material="ur5_mat"/>

                    <body name="upper_arm_link" pos="0 0.13585 0" quat="0.707107 0 0.707107 0">
                        <inertial pos="0 0 0.28" mass="8.393" diaginertia="0.226891 0.226891 0.0151074" />
                        <joint name="shoulder_lift_joint" class="UR5" pos="0 0 0" axis="0 1 0" limited="true" range="-6.28318 6.28318" /> <!--Range= -6.28318 6.28318-->
                        <geom name="upperarm_geom" type="mesh"  mesh="upperarm" material="ur5_mat" />

                        <body name="forearm_link" pos="0 -0.1197 0.425">
                            <inertial pos="0 0 0.25" mass="2.275" diaginertia="0.0494433 0.0494433 0.004095" />
                            <joint name="elbow_joint" class="UR5" pos="0 0 0" axis="0 1 0" limited="true" range="-6.28318 6.28318" /> <!--Range= -6.28318 6.28318-->
                            <geom name="forearm_geom" type="mesh"  mesh="forearm" material="ur5_mat"/>

                            <body name="wrist_1_link" pos="0 0 0.39225" quat="0.707107 0 0.707107 0"> <!-- !! quat="1 0 1 0" !!quat="0.707107 0 0.707107 0" > <!-quat="0.707107 0
                                <inertial pos="0 0 0" quat="0.5 0.5 -0.5 0.5" mass="1.219" diaginertia="0.21942 0.111173 0.111173" />
                                <joint name="wrist_1_joint" class="UR5e" pos="0 0 0" axis="0 1 0" limited="true" range="-6.28318 6.28318" />
                                <geom name="wrist_1_geom" type="mesh"  mesh="wrist1" material="ur5_mat"/>
```

Fig. 5.5: Kinematic structure of the robot model used in the simulation environment.

5.3.3 Gripper Modelling

The end-effector used in the simulation is a parallel-jaw gripper (Figure 5.6), which is one of the most common gripper configurations used in industrial robotic manipulation tasks. Parallel-jaw grippers consist of two opposing fingers that move symmetrically toward or away from each other to grasp objects.

In the MuJoCo simulation model, the gripper is represented as a pair of rigid bodies connected by prismatic joints that allow linear motion along the closing direction of the gripper. These joints are controlled by actuators that apply forces to move the gripper fingers during the grasping operation. The contact surfaces of the gripper fingers are defined using geometric primitives that interact with the object geometry during the simulation.

The gripping force applied by the gripper is determined based on the analytical model described in Chapter 4. The calculated gripping force is applied through the gripper actuators during the simulation, allowing the stability of the grasp to be evaluated under realistic physical conditions. The friction coefficient between the gripper fingers and the object surface is also defined within the simulation model to represent the contact interaction accurately.

By modelling the gripper and object interactions in this way, the simulation environment can reproduce the mechanical behaviour of the grasp and verify whether the analytically calculated parameters are sufficient to maintain a stable grasp throughout the manipulation process.

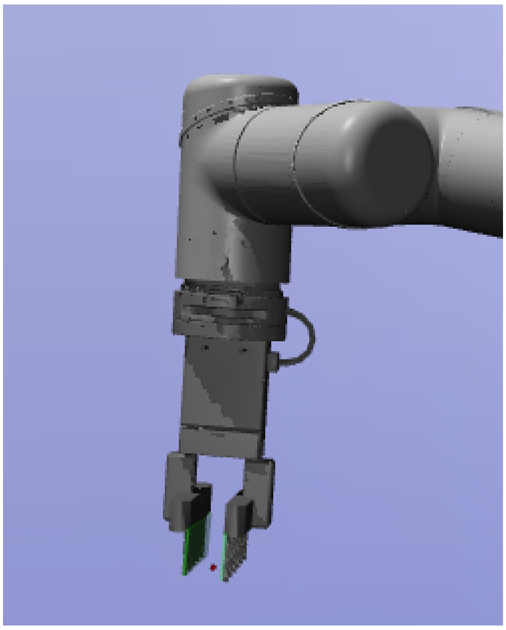

Fig. 5.6: Parallel-jaw gripper model used in the MuJoCo simulation environment.

5.4 Experimental Design

5.4.1 Experimental Objectives

The primary objective of the experimental evaluation is to validate the proposed framework for automated gripping parameter estimation. The framework combines CAD-based geometric analysis, analytical gripping force calculation, and physics-based simulation to generate and validate parameters required for robotic pick-and-place operations.

The experiments are designed to evaluate whether the parameters generated by the analytical model are sufficient to ensure stable object manipulation under realistic physical conditions. In particular, the evaluation focuses on the following aspects:

- correctness of the estimated gripping force
- feasibility of the computed gripping location
- stability of the grasp during object lifting and transportation

By executing the manipulation tasks within the MuJoCo simulation environment, the behaviour of the robotic system can be observed and analyzed under physically realistic conditions. Physics-based simulation has become an important tool in robotic manipulation research because it allows rapid testing and validation of manipulation strategies before deployment in real robotic systems [7].

The experiments therefore aim to demonstrate that the proposed CAD-based parameter estimation method can generate gripping parameters that are suitable for stable robotic manipulation.

5.4.2 Experimental Setup

The experimental evaluation was conducted using the simulation architecture described in the previous sections. The complete workflow integrates CAD geometry processing, analytical parameter estimation, structured data representation, and physics-based robotic simulation.

The experimental setup consists of the following main components:

1. ***CAD Processing Environment***

FreeCAD is used to process the object CAD model and extract geometric parameters such as object dimensions, bounding box size, and center of mass location.

2. ***Analytical Parameter Estimation Module***

The analytical gripping model developed in Chapter 4 calculates the required gripping force and determines suitable gripping locations based on the extracted geometric and physical parameters.

3. ***Parameter Export***

The calculated parameters are stored and exported as a structured JSON file that serves as the interface between the parameter estimation module and the simulation environment.

4. ***Simulation Environment***

The MuJoCo physics engine is used to simulate the robotic pick-and-place task using the generated gripping parameters.

5. ***Control System***

A Python-based control script reads the JSON parameter file containing the grasp candidates and executes the manipulation sequence within the simulation environment. Only the candidate with the minimum value among all the other feasible candidates is selected as the optimized parameter.

The integration of these components forms a complete pipeline (Figure 5.7) for automated gripping parameter estimation and validation.

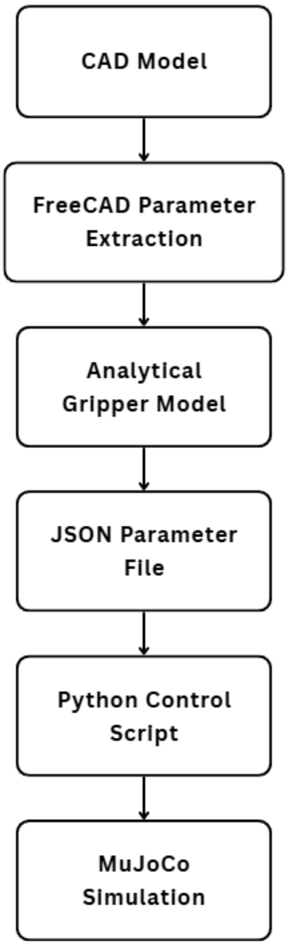

Fig. 5.7: Experimental framework for automated gripping parameter estimation and simulation validation.

5.4.3 Test Objects

To evaluate the proposed parameter estimation method, several object models were used in the experiments. The objects were selected to represent typical industrial components that may be handled by robotic manipulation systems.

The test objects include:

- cylindrical battery cell
- rectangular battery module
- simple box-shaped component

These objects were chosen because they represent common geometries encountered in automated assembly environments and allow the evaluation of gripping parameter estimation for different object shapes.

The CAD models of the objects were imported into the FreeCAD environment, where the geometric properties required for the analytical calculations were extracted.

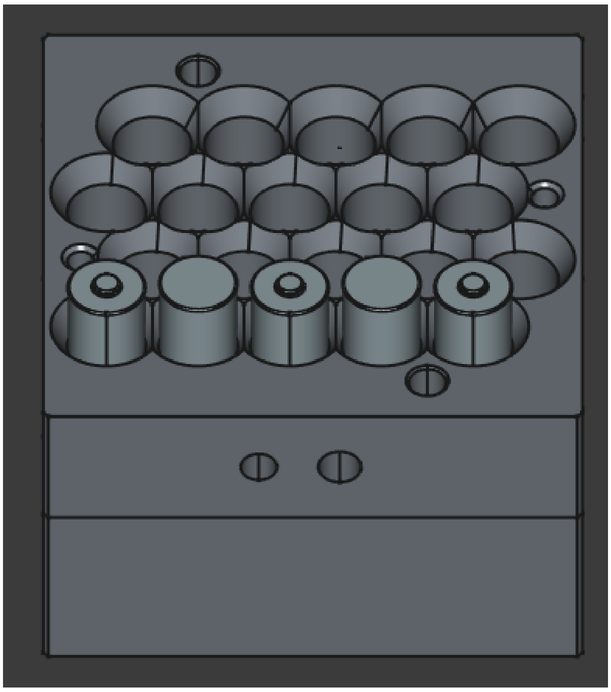

Fig. 5.8: CAD models of the test objects used in the experimental evaluation.

6 Discussion

6.1 Analysis and Interpretation of Results

The experimental results presented in Chapter 5 demonstrate the feasibility of the proposed framework for automated gripping parameter estimation. By integrating CAD-based geometric analysis, analytical gripping force estimation, and physics-based simulation, the framework enables the automatic generation and validation of parameters required for robotic pick-and-place operations. The results obtained from the simulation experiments provide insight into the effectiveness of the analytical model and the overall system architecture.

One of the key observations from the experimental evaluation is that the gripping parameters generated by the analytical model were sufficient to maintain stable object manipulation across the tested scenarios. The gripping forces calculated using the mechanical model described in Chapter 4 successfully prevented slippage during the lifting and transportation phases of the manipulation task. This indicates that the analytical formulation based on frictional contact constraints and torque equilibrium provides a reasonable approximation of the forces required for stable grasping under the assumed conditions.

The successful manipulation of the object geometries further demonstrates the flexibility of the CAD-based parameter extraction approach. Because the geometric parameters are obtained directly from the CAD model, the framework can adapt to different object shapes with a minimal manual parameter tuning. This capability is particularly relevant in automated manufacturing environments, where new product variants must often be integrated into existing production systems with minimal engineering effort. CAD-based manipulation planning approaches have been identified as a promising strategy for enabling such flexible automation systems[4].

Another important aspect of the results is the integration between the analytical parameter estimation process and the simulation environment. The use of a structured JSON interface allowed the parameters generated by the analytical model to be seamlessly transferred to the MuJoCo simulation environment. This modular architecture supports the separation of parameter generation and simulation validation, enabling independent development of the different components within the system framework.

The simulation results also demonstrate the importance of incorporating realistic physical modelling when evaluating robotic manipulation strategies. By modelling rigid-body dynamics and contact interactions within the MuJoCo physics engine, it was possible to observe the behaviour of the grasped object under realistic conditions. This includes the influence of friction, contact forces, and object inertia on the stability of the manipulation task. Simulation environments capable of modelling such physical interactions are increasingly used in robotics research to evaluate manipulation strategies before deploying them on physical robotic systems [7].

From a methodological perspective, the results indicate that the combination of analytical modelling and physics-based simulation provides an effective approach for developing robotic manipulation strategies. Analytical models allow the rapid estimation of gripping parameters without the computational complexity associated with large-scale grasp planning algorithms. At the same time, simulation environments provide a mechanism for validating the analytical predictions under realistic physical conditions.

The experimental results also highlight the importance of accurate geometric and physical parameter extraction. Since the analytical model relies on parameters such as object mass, center of mass location, and friction coefficient, inaccuracies in these values may affect the resulting gripping force estimation. In the experiments conducted in this thesis, the use of CAD models allowed these parameters to be determined with sufficient accuracy to enable stable manipulation. However, in practical applications involving uncertain object properties, additional sensing or adaptive control strategies may be required to ensure reliable manipulation.

Overall, the results of the experimental evaluation demonstrate that the proposed approach provides a viable method for generating gripping parameters directly from digital product models. The integration of CAD-based geometry processing, analytical force estimation, and simulation-based validation enables the automated generation of manipulation parameters that can be used for robotic pick-and-place operations. This approach contributes to the broader objective of increasing the level of automation in robotic manipulation planning by reducing the need for manual parameter tuning and trial-and-error experimentation.

However, while the simulation results confirm the feasibility of the proposed framework under controlled conditions, further investigation is required to evaluate its performance in real-world robotic systems. In practical applications, additional factors such as sensor noise, mechanical compliance, and uncertainties in material properties may influence the behaviour of the manipulation process. These aspects are therefore considered in the following section, which discusses the limitations of the proposed approach and its implications for industrial applications.

6.2 Methodological Limitations and Industrial Implications

While the experimental results presented in Chapter 5 demonstrate the feasibility of the proposed framework for automated gripping parameter estimation, several limitations of the methodology must be considered when evaluating its applicability to real-world robotic systems. These limitations primarily arise from the assumptions made in the analytical modelling process, the simplifications inherent in the simulation environment, and the constraints associated with CAD-based parameter extraction.

One of the primary limitations of the proposed approach is the reliance on simplified analytical models for estimating the required gripping force. The analytical formulation used in this thesis assumes quasi-static conditions and relies on classical friction-based contact models to determine the minimum gripping force required to prevent slippage. While such models are widely used in robotic grasp analysis and provide useful first-order approximations of grasp stability, they do not fully capture the complex interactions that may occur in real manipulation scenarios. For example, variations in surface roughness, deformation of the contact surfaces, and uncertainties in friction coefficients can significantly affect the actual contact forces experienced during grasping operations [5].

Another limitation arises from the assumption that the geometric and physical properties of the manipulated object can be accurately extracted from CAD models. In many industrial settings, CAD models provide precise descriptions of object geometry, which makes them suitable for geometric parameter extraction. However, the actual physical properties of manufactured objects may differ from their digital representations due to manufacturing tolerances, material inconsistencies, or assembly variations. These deviations may lead to inaccuracies in the calculated gripping parameters, particularly in cases where the center of mass location or object mass differs from the values assumed in the model.

The proposed framework also assumes that the friction coefficient between the gripper fingers and the object surface is known and remains constant during manipulation. In practice, friction coefficients are influenced by several factors, including surface contamination, wear of the contact surfaces, and environmental conditions such as temperature or humidity. As a result, the actual friction conditions during manipulation may differ from the assumed values used in the analytical calculations. In industrial systems, such uncertainties are often addressed by introducing safety factors in the gripping force estimation or by incorporating sensor-based feedback mechanisms to monitor grasp stability.

Another important limitation relates to the use of simulation-based validation rather than physical experimentation. Although physics engines such as MuJoCo provide accurate modelling of rigid-body dynamics and contact interactions, simulation environments cannot perfectly reproduce all aspects of real-world robotic manipulation. Factors such as mechanical compliance, actuator dynamics, sensor noise, and hardware imperfections may influence the behaviour of real robotic systems in ways that are not fully captured by simulation models [7]. Consequently, while the simulation results obtained in this thesis provide valuable insight into the feasibility of the proposed framework, further validation using physical robotic platforms would be necessary to fully assess the robustness of the method in real industrial environments.

In summary, although the proposed framework is subject to several methodological limitations, the results obtained in this thesis demonstrate its potential as a practical tool for generating robotic manipulation parameters directly from digital product models. The integration of CAD-based geometry analysis, analytical gripping force estimation, and physics-based simulation provides a coherent workflow for developing and validating robotic pick-and-place strategies. Future research may further enhance the framework by incorporating real-time perception systems, adaptive control strategies, and experimental validation on physical robotic platforms.

7 Summary and Outlook

7.1 Summary of Contributions

This thesis addressed the problem of automatically determining gripping parameters for robotic pick-and-place operations based on digital object models. In modern manufacturing environments, robotic systems must frequently adapt to new products and production scenarios. Traditional approaches often rely on manual parameter tuning, extensive trial-and-error experimentation, or complex data-driven grasp planning methods. These approaches can require significant engineering effort and may limit the flexibility of automated production systems. The objective of this research was therefore to develop a systematic framework that enables the automated generation and validation of gripping parameters using digital product information.

The proposed framework integrates CAD-based geometry processing, analytical gripping force estimation, structured data representation, and physics-based simulation to generate and validate manipulation parameters for robotic pick-and-place tasks. By combining these components within a unified workflow, the framework enables the automated estimation of gripping parameters directly from digital object models.

The first major contribution of this thesis is the development of a CAD-driven parameter extraction approach for robotic manipulation planning. Using the FreeCAD environment, geometric properties such as object dimensions, bounding box parameters, and center-of-mass locations can be automatically extracted from CAD models. These parameters provide the essential input data required for analytical gripping calculations and eliminate the need for manual parameter specification.

The second contribution is the formulation of an analytical gripping parameter estimation method based on classical principles of rigid-body mechanics and frictional contact theory. The analytical model determines the minimum gripping force required to maintain grasp stability by considering gravitational forces, friction constraints, and torque equilibrium conditions. This approach provides a computationally efficient alternative to large-scale grasp planning algorithms while maintaining physical interpretability and transparency.

The third contribution is the implementation of a data interface for transferring manipulation parameters between the analytical model and the simulation environment. The use of a structured JSON-based representation enables the seamless integration of parameter estimation modules with robotic control software and simulation tools.

Finally, the framework was validated through physics-based simulation experiments using the MuJoCo environment. The simulation experiments demonstrated that the gripping parameters generated by the analytical model were sufficient to perform stable pick-and-place operations across several test scenarios. The integration of analytical modelling and physics-based simulation provides an effective mechanism for validating manipulation strategies before deployment in real robotic systems.

Overall, the results of this thesis demonstrate that it is possible to generate robotic gripping parameters directly from digital product models by combining CAD-based geometry processing, analytical mechanical modelling, and simulation-based validation. This approach contributes to the broader goal of increasing the level of automation in robotic manipulation planning and reducing the engineering effort required for configuring robotic pick-and-place systems.

7.2 Future Research Directions

Although the results obtained in this thesis demonstrate the feasibility of the proposed framework, several opportunities for further research and development remain.

One potential direction for future work is the integration of perception-based object recognition and pose estimation methods. While the framework developed in this thesis relies on CAD models for parameter extraction, many real-world applications involve partially unknown object poses or dynamic environments. Integrating vision-based perception systems could enable the framework to operate in more flexible and unstructured environments by combining geometric knowledge from CAD models with real-time sensor data.

Another important area for future research is the extension of the analytical model to include more complex contact interactions and object properties. The analytical model used in this thesis assumes rigid-body dynamics and simplified frictional contact conditions. However, real-world manipulation tasks may involve deformable objects, compliant gripper surfaces, or complex contact geometries. Extending the analytical model to account for these factors could improve the accuracy of gripping parameter estimation in more challenging manipulation scenarios.

Future work could also explore the integration of machine learning techniques with the analytical modelling approach. Data-driven methods have shown promising results in robotic grasp planning, particularly for complex object geometries. Combining analytical models with learning-based approaches may enable the development of hybrid manipulation planning systems that leverage both physical knowledge and data-driven optimization.

Another promising research direction is the implementation and validation of the proposed framework on a physical robotic system. While the simulation experiments performed in this thesis provide valuable insight into the feasibility of the approach, real-world experiments would allow the evaluation of additional factors such as actuator dynamics, sensor noise, and mechanical compliance. Experimental validation on industrial robotic platforms would therefore represent an important step toward practical deployment of the proposed method.

Finally, further research could investigate the integration of the framework within digital twin environments for manufacturing systems. Digital twins provide virtual representations of physical production systems and enable advanced simulation and optimization capabilities. Integrating the proposed gripping parameter estimation method within digital twin platforms could support automated planning and validation of robotic manipulation tasks in complex production environments.

In conclusion, the framework developed in this thesis provides a foundation for automated gripping parameter estimation based on digital product models. By combining analytical modelling, CAD-based parameter extraction, and simulation-based validation, the proposed approach contributes to the development of more flexible and autonomous robotic manipulation systems. Continued research in this direction has the potential to further enhance the capabilities of robotic automation and support the realization of adaptive and intelligent manufacturing systems.

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